

Grade 12  
Pre-Calculus Mathematics  
Achievement Test

# **Marking Guide**

June 2018

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While the department is committed to making its publications as accessible as possible, some parts of this document are not fully accessible at this time.

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# General Marking Instructions

**Please do not make any marks in the student test booklets.** If the booklets have marks in them, the marks will need to be removed by departmental staff prior to sample marking should the booklet be selected.

Please ensure that

- the booklet number and the number on the *Answer/Scoring Sheet* are identical
- **students and markers use only a pencil to complete the *Answer/Scoring Sheets***
- the totals of each of the four parts are written at the bottom
- each student's final result is recorded, by booklet number, on the corresponding *Answer/Scoring Sheet*
- the *Answer/Scoring Sheet* is complete
- a photocopy has been made for school records

Once marking is completed, please forward the *Answer/Scoring Sheets* to Manitoba Education and Training in the envelope provided (for more information see the administration manual).

## Marking the Test Questions

The test is composed of constructed response questions and selected response questions. Constructed response questions are worth 1 to 5 marks each, and selected response questions are worth 1 mark each. An answer key for the selected response questions can be found at the beginning of the section "Booklet 2 Questions."

To receive full marks, a student's response must be complete and correct. Where alternative answering methods are possible, the *Marking Guide* attempts to address the most common solutions. For general guidelines regarding the scoring of students' responses, see Appendix A.

## Irregularities in Provincial Tests

During the administration of provincial tests, supervising teachers may encounter irregularities. Markers may also encounter irregularities during local marking sessions. Appendix B provides examples of such irregularities as well as procedures to follow to report irregularities.

If an *Answer/Scoring Sheet* is marked with "0" and/or "NR" only (e.g., student was present but did not attempt any questions), please document this on the *Irregular Test Booklet Report*.

## **Assistance**

If, during marking, any marking issue arises that cannot be resolved locally, please call Manitoba Education and Training at the earliest opportunity to advise us of the situation and seek assistance if necessary.

You must contact the Assessment Consultant responsible for this project before making any modifications to the answer keys or scoring rubrics.

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## Communication Errors

The marks allocated to questions are primarily based on the concepts and procedures associated with the learning outcomes in the curriculum. For each question, shade in the circle on the *Answer/Scoring Sheet* that represents the marks given based on the concepts and procedures. A total of these marks will provide the preliminary mark.

Errors that are not related to concepts or procedures are called “Communication Errors” (see Appendix A) and will be tracked on the *Answer/Scoring Sheet* in a separate section. There is a  $\frac{1}{2}$  mark deduction for each type of communication error committed, regardless of the number of errors per type (i.e., committing a second error for any type will not further affect a student’s mark), with a maximum deduction of 5 marks from the total test mark.

When a given response includes multiple types of communication errors, deductions are indicated in the order in which the errors occur in the response. No communication errors are recorded for work that has not been awarded marks. The total deduction may not exceed the marks awarded.

The student’s final mark is determined by subtracting the communication errors from the preliminary mark.

Example: A student has a preliminary mark of 72. The student committed two E1 errors ( $\frac{1}{2}$  mark deduction), four E7 errors ( $\frac{1}{2}$  mark deduction), and one E8 error ( $\frac{1}{2}$  mark deduction). Although seven communication errors were committed in total, there is a deduction of only  $1\frac{1}{2}$  marks.

| COMMUNICATION ERRORS / ERREURS DE COMMUNICATION                                                                                                                                                                                      |   |    |   |    |   |    |   |     |   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|----|---|----|---|-----|---|
| Shade in the circles below for a maximum total deduction of 5 marks ( $\frac{1}{2}$ mark deduction per error).<br>Noircir les cercles ci-dessous pour une déduction maximale totale de 5 points (déduction de 0,5 point par erreur). |   |    |   |    |   |    |   |     |   |
| E1                                                                                                                                                                                                                                   | ● | E2 | ○ | E3 | ○ | E4 | ○ | E5  | ○ |
| E6                                                                                                                                                                                                                                   | ○ | E7 | ● | E8 | ● | E9 | ○ | E10 | ○ |

Example: Marks assigned to the student.

| Marks Awarded      | Booklet 1 | Selected Response | Booklet 2 | Communication Errors (Deduct)       | Total           |
|--------------------|-----------|-------------------|-----------|-------------------------------------|-----------------|
|                    | 25        | 7                 | 40        | $1\frac{1}{2}$                      | $70\frac{1}{2}$ |
| <b>Total Marks</b> | <b>36</b> | <b>9</b>          | <b>45</b> | <b>maximum deduction of 5 marks</b> | <b>90</b>       |



# Scoring Guidelines for Booklet 1 Questions

---



## Question 1

T1

Pierre pushes his car into a garage. The radius of a tire on his car is 22 cm. Determine the distance travelled by his car if the tire rotated a total of  $1000^\circ$ .

### Solution

$$\theta = (1000) \left( \frac{\pi}{180} \right)$$

1 mark for conversion

$$= \frac{50\pi}{9}$$

$$s = \theta r$$

$$s = \left( \frac{50\pi}{9} \right) (22)$$

1 mark for substitution

$$= \frac{1100\pi}{9} \text{ cm}$$

**2 marks**

**or**

$$s = 383.972 \text{ cm}$$

### Exemplar 1

---

$$S = \theta r \quad \theta = 1000 \times \frac{\pi}{180} \approx 17.4533$$

$$S = \left(\frac{1000\pi}{180}\right)(22)$$

$$S \approx 34.907$$

---

**1½ out of 2**

award full marks

– ½ mark for arithmetic error in line 3

E5 (units of measure omitted in final answer)

### Exemplar 2

---

$$S = \theta r$$

$$S = 1000(22)$$

$$S = \underline{22000} \text{ cm} \quad \text{or } 220 \text{ m}$$

---

**1 out of 2**

+ 1 mark for substitution

### Exemplar 3

---

$$r = 11 \text{ cm}$$

$$S = \theta r$$

$$S = \frac{1000\pi}{180} \cdot 11$$

$$S = 191.986 \text{ cm}$$

---

**1½ out of 2**

award full marks

– ½ mark for procedural error in line 1

Solve, algebraically.

$$7^{\frac{x}{2}} = 85$$

**Solution**

$$\log 7^{\frac{x}{2}} = \log 85$$

½ mark for applying logarithms

$$\frac{x}{2} \log 7 = \log 85$$

1 mark for power law

$$x \log 7 = 2 \log 85$$

$$x = \frac{2 \log 85}{\log 7}$$

$$x = 4.566\ 142$$

½ mark for evaluating quotient of logarithms

$$x = 4.566$$

**2 marks**

### Exemplar 1

---

$$\log_7 85 = \frac{x}{2}$$

$$\frac{\log 85}{\log 7} = \frac{x}{2}$$

$$x = 2 \left( \frac{\log 85}{\log 7} \right)$$

$$x = 4.566$$

---

2 out of 2

### Exemplar 2

---

$$\log 7^{\frac{x}{2}} = \log 85$$

$$\left(\frac{x}{2}\right) \log 7 = \log 85$$

$$x \log 7 - 2 \log 7 = \log 85$$

$$x \log 7 = \log 85 + 2 \log 7$$

$$x = \frac{\log 85 + 2 \log 7}{\log 7}$$

$$x = 4.283$$

---

1 out of 2

award full marks

– 1 mark for concept error in line 3

Solve, algebraically, over the interval  $[0, 2\pi)$ .

$$\sin x (\sec x + 3) = 0$$

**Solution**

$$\sin x = 0$$

$$\sec x = -3$$

$\frac{1}{2}$  mark for solving for  $\sin x$

$\frac{1}{2}$  mark for solving for  $\sec x$

$$\cos x = -\frac{1}{3}$$

1 mark for reciprocal

$$x_r = 1.230\ 959$$

$$x = 0, \pi$$

$$x = 1.911, 4.373$$

2 marks for values of  $x$  (1 mark for each branch)

**4 marks**

## Exemplar 1

---

$$\sin x (\sec x + 3) = 0$$

$$\sin x = 0 \quad \sec x = -3$$

$$x = 0, \pi, 2\pi \quad \cos x = -\frac{1}{3}$$

$$\theta = 0.944956946$$

$$\text{QII: } x = \pi - \theta$$

$$x = \pi - 0.944956946$$

$$x = 2.197$$

$$\text{QIII: } x = \pi + \theta$$

$$x = \pi + 0.944956946$$

$$x = 4.087$$

$$x = 0, 2.197, \pi, 4.087, 2\pi$$

---

**3½ out of 4**

award full marks

– ½ mark for procedural error in line 4

E3 (variable introduced without being defined in line 4)

E8 (answer outside the given domain)

## Exemplar 2

---

$$\sin x \left( \frac{1}{\cos x} + 3 \right) = 0$$

$$\sin x = 0$$

$$x = 0$$

$$\frac{1}{\cos x} + 3 = 0$$

$$\cos x = -3$$

NO SOLUTION

---

**3 out of 4**

+ 1 mark for reciprocal

+ ½ mark for solving for  $\sin x$

+ ½ mark for value of  $x$

+ 1 mark for no solution (consistent with error)

## Exemplar 3

---

$$\sin x \left( \frac{1}{\sin x} + 3 \right) = 0$$

$$\sin x = 0 \quad \sin x = -\frac{1}{3}$$

$$0, \pi, \cancel{2\pi}$$

$$x_R = 0.3398$$

$$x = 3.481$$

$$x = 5.943$$

---

**3 out of 4**

award full marks

– 1 mark for concept error in line 1

E7 (notation error in line 3)

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## Question 4

R10

Brahim invests \$2500 at an annual interest rate of 6.75% compounded monthly. Determine, algebraically, how many years it will take for his investment to reach an amount of \$10 500.

Use the formula:

$$A = P \left( 1 + \frac{r}{n} \right)^{nt}$$

where  $A$  = the amount of the investment after  $t$  years

$P$  = the principal of the investment

$r$  = the annual interest rate (as a decimal)

$n$  = the number of compounding periods per year

$t$  = the length of the investment in years

### Solution

$$10\,500 = 2500 \left( 1 + \frac{0.0675}{12} \right)^{12t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$4.2 = (1.005\,625)^{12t} \quad \frac{1}{2} \text{ mark for simplification}$$

$$\log 4.2 = 12t \log 1.005\,625 \quad \frac{1}{2} \text{ mark for applying logarithms}$$

$$\frac{\log 4.2}{12 \log 1.005\,625} = t \quad 1 \text{ mark for power law}$$

$$21.320\,250 = t$$

$$21.320 \text{ years} = t$$

$\frac{1}{2}$  mark for evaluating quotient of logarithms

**3 marks**

## Exemplar 1

---

$$A = 10\,500$$

$$P = 2\,500$$

$$r = 6.75$$

$$n = 12$$

$$t = ?$$

$$A = P \left( 1 + \frac{r}{n} \right)^{nt}$$

$$\frac{10\,500}{2\,500} = \frac{2\,500 \left( 1 + \frac{(6.75)}{(12)} \right)^{(12)t}}{2\,500}$$

$$4.2 = 1 + \left( \frac{6.75}{12} \right)^{12t}$$

$$4.2 = 1 + (0.5625)^{12t}$$

$$4.2 = (1.5625)^{12t}$$

$$\log 4.2 = \log 1.5625^{12t}$$

$$\frac{\log 4.2}{\log 1.5625} = \frac{12t + \cancel{\log 1.5625}}{\cancel{\log 1.5625}}$$

$$t = 0.268 \text{ years}$$

---

**2½ out of 3**

+ ½ mark for simplification

+ ½ mark for applying logarithms

+ 1 mark for power law

+ ½ mark for evaluating quotient of logarithms

E4 (missing brackets but still implied in lines 3 and 4)

## Exemplar 2

---

$$10\,500 = 2500 \left(1 + \frac{0.0675}{12}\right)^{(12)t} \quad r = 0.0675$$

$$\frac{10500}{2500} = \frac{2500}{2500} (1.005625)^{12t}$$

$$4.2 = 1.005625^{12t}$$
$$\log 4.2 = \log 1.005625^{12t}$$
$$\log 4.2 = 12t \log 1.005625$$

$\therefore$  It would take 2 years

$$0.623144190 = 0.285200663t$$

$$t = 2.185$$

---

### 3 out of 3

award full marks

E7 (transcription error in line 1)

E6 (rounding error in final answer)

## Exemplar 3

---

$$10,500 = 2500 \left(1 + \frac{0.0675}{12}\right)^{12t}$$

$$\frac{10,500}{2500} = \frac{2500}{2500} (1.005625)^{12t}$$

$$4.2 = 1.005625^{12t}$$

$$\log 4.2 = 12t \log 1.005625$$
$$\frac{\log 4.2}{\log 1.005625}$$

---

### 2½ out of 3

+ ½ mark for substitution

+ ½ mark for simplification

+ ½ mark for applying logarithms

+ 1 mark for power law

E7 (notation error in lines 2 and 4)

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There are 13 adults and 18 children who can be selected to go on a trip. Determine the number of ways 4 adults and 7 children can be selected if Sandra, one of the adults, must be selected.

**Solution**

$${}_1C_1 \cdot {}_{12}C_3 \cdot {}_{18}C_7$$

1 mark for  ${}_{12}C_3$

$\frac{1}{2}$  mark for  ${}_{18}C_7$

7 001 280

$\frac{1}{2}$  mark for product of combinations

**2 marks**

---

Note:

- ${}_1C_1$  does not need to be shown.

## Exemplar 1

---

$${}_{12}C_4 \times {}_{18}C_7 \times {}_1C_1 = 15\,752\,860$$

### 1 out of 2

+ ½ mark for  ${}_{18}C_7$

+ ½ mark for product of combinations

## Exemplar 2

---

$$\begin{array}{cc} {}_{12}C_3 & {}_{18}C_7 \\ 220 & 31824 \end{array} = 32044 \text{ ways}$$

### 1½ out of 2

+ 1 mark for  ${}_{12}C_3$

+ ½ mark for  ${}_{18}C_7$

## Exemplar 3

---

$$\begin{array}{c} \underline{1} \cdot \underline{12} \cdot \underline{11} \cdot \underline{10} \\ \text{Seneca} \end{array} \times \underline{18} \cdot \underline{17} \cdot \underline{16} \cdot \underline{15} \cdot \underline{14} \cdot \underline{13} \cdot \underline{12} =$$
$$= 2,117 \times 10^{11} \text{ ways}$$

### 1 out of 2

award full marks

– 1 mark for concept error (using permutations instead of combinations)

In the binomial expansion of  $\left(\frac{2}{x^2} - x^3\right)^9$ , determine and simplify the 6<sup>th</sup> term.

**Solution**

$$t_6 = {}_9C_5 \left(\frac{2}{x^2}\right)^4 (-x^3)^5 \quad 2 \text{ marks (1 mark for } {}_9C_5; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$t_6 = 126 \left(\frac{16}{x^8}\right) (-x^{15})$$

$$t_6 = -2016x^7 \quad 1 \text{ mark for simplification (}\frac{1}{2}\text{ mark for coefficient; } \frac{1}{2}\text{ mark for exponent)}$$

**3 marks**

## Exemplar 1

---

$$t_{k+1} = nC_k x^{n-k} y^k$$

$$t_{5+1} = (9C_5) \left(\frac{2}{x^2}\right)^{9-5} (-y^3)^5$$

$$t_6 = (126) \left(\frac{16}{x^4}\right) (-x^{13})$$

$$t_6 = \frac{-2016 x^{15}}{x^4}$$

$$t_6 = -2016 x^{11}$$

---

**2½ out of 3**

+ 1 mark for  $9C_5$

+ 1 mark for consistent factors

+ ½ mark for simplification of coefficient

## Exemplar 2

---

$$t_6 = t_{k+1}$$

$$k = 5$$

$$t_6 = 9C_6 \cdot \left(\frac{2}{x^2}\right)^3 \cdot (-x^3)^6$$

$$t_6 = \frac{9!}{3!6!} \cdot \frac{8}{x^6} \cdot -x^{18}$$

$$t_6 = \frac{9 \cdot 8 \cdot 7 \cdot 6!}{3! \cdot 6!} \cdot \frac{-8 \times 18}{x^6}$$

$$t_6 = 84 \cdot -8 \times 12$$

$$t_6 = -672 x^{12}$$

---

**1½ out of 3**

+ 1 mark for consistent factors

+ ½ mark for simplification of exponent

Given that  $f(x) = \{(-1, 0), (0, 2), (1, -3), (2, 4)\}$ , evaluate  $f(f(0))$ .

**Solution**

$$f(f(0))$$

$$f(2)$$

$$4$$

$\frac{1}{2}$  mark for  $f(0) = 2$

$\frac{1}{2}$  mark for  $f(f(0))$

**1 mark**

### Exemplar 1

---

$$f(0) = -1$$

$$f(-1) = 0$$

---

**½ out of 1**

+ ½ mark for  $f(f(0))$  (consistent with error)

### Exemplar 2

---

$$f(f(0))$$

$$f(2)$$

---

**½ out of 1**

+ ½ mark for  $f(0) = 2$

The point  $\left(-\frac{5}{6}, b\right)$  is on the unit circle and is in quadrant III.

Determine the exact value of  $b$ .

**Solution**

$$\left(-\frac{5}{6}\right)^2 + b^2 = 1 \quad \frac{1}{2} \text{ mark for substitution}$$

$$b^2 = 1 - \frac{25}{36}$$

$$b^2 = \frac{11}{36}$$

$$b = \pm \frac{\sqrt{11}}{6}$$

$$b = -\frac{\sqrt{11}}{6}$$

$\frac{1}{2}$  mark for the exact value of  $b$

**1 mark**

## Exemplar 1

---

$$x = -5 \quad r = 6 \quad y = -\sqrt{11}$$

$$x^2 + y^2 = r^2$$

$$y^2 = r^2 - x^2$$

$$y^2 = 36 - 25$$

$$y^2 = 11$$

$$y = \pm\sqrt{11}$$

$$P(\theta) = \left(-\frac{5}{6}, -\frac{\sqrt{11}}{6}\right)$$

---

**1 out of 1**

## Exemplar 2

---

$$r^2 = x^2 + y^2$$

$$1^2 = \left(-\frac{5}{6}\right)^2 + b^2$$

$$1 = \frac{25}{36} + b^2$$

$$1 - \frac{25}{36} = b^2$$

$$\sqrt{\frac{11}{36}} = \sqrt{b^2}$$

$$\frac{\sqrt{11}}{6} = b$$

---

**½ out of 1**

+ ½ mark for substitution

### Exemplar 3

---

$$\begin{aligned}x^2 + y^2 &= 1 \\y^2 &= 1 - x^2 \\y^2 &= 1 - \left(-\frac{5}{6}\right)^2 \\&= 1 - \frac{25}{36} \\&= \frac{36}{36} - \frac{25}{36} \\&= \frac{11}{36} \\\therefore y &= \pm \frac{\sqrt{11}}{6} \\y &= -\frac{\sqrt{11}}{6}\end{aligned}$$

---

**1 out of 1**

award full marks

E1 (final answer not stated)

### Exemplar 4

---

$$\begin{aligned}\left(-\frac{5}{6}\right)^2 + b^2 &= 1 \\-\frac{25}{36} + b^2 &= 1 \\b^2 &= 1 + \frac{25}{36} \\b &= -\frac{\sqrt{61}}{6}\end{aligned}$$

---

**½ out of 1**

award full marks

– ½ mark for arithmetic error in line 2

E1 (impossible solution not rejected in final answer)

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### Question 9

P4

Given the following row of Pascal's Triangle, determine the values of the next row.

1   6   15   20   15   6   1

#### **Solution**

1   7   21   35   35   21   7   1

**1 mark**

### Exemplar 1

---

7 21 35 35 21 7

---

**0 out of 1**

### Exemplar 2

---

1 7 21 35 21 7 1

---

**0 out of 1**

## Question 10

R3, R5

The following transformations are applied to  $f(x)$ , resulting in a new function,  $g(x)$ .

- reflection over the  $x$ -axis
- vertical stretch by a factor of 3
- horizontal stretch by a factor of 4

State the equation of  $g(x)$  in terms of  $f(x)$ .

### Solution

$$\underline{g(x) = -3f\left(\frac{1}{4}x\right)}$$

1 mark for vertical reflection

1 mark for vertical stretch

1 mark for horizontal stretch

**3 marks**

## Exemplar 1

---

$$g(x) = \underline{3 - f(4x)}$$

**1 out of 3**

+ 1 mark for vertical reflection

## Exemplar 2

---

$$g(x) = \underline{3g(-\frac{1}{4}x)}$$

**1½ out of 3**

+ 1 mark for vertical stretch

+ 1 mark for horizontal stretch

– ½ mark for procedural error ( $g$  instead of  $f$ )

## Exemplar 3

---

$$f(x) = a(b(x-h)) + k$$

$$a = -3$$

$$b = \frac{1}{4}$$

$$g(x) = \underline{-3\left(\frac{1}{4}(x)\right)}$$

**2 out of 3**

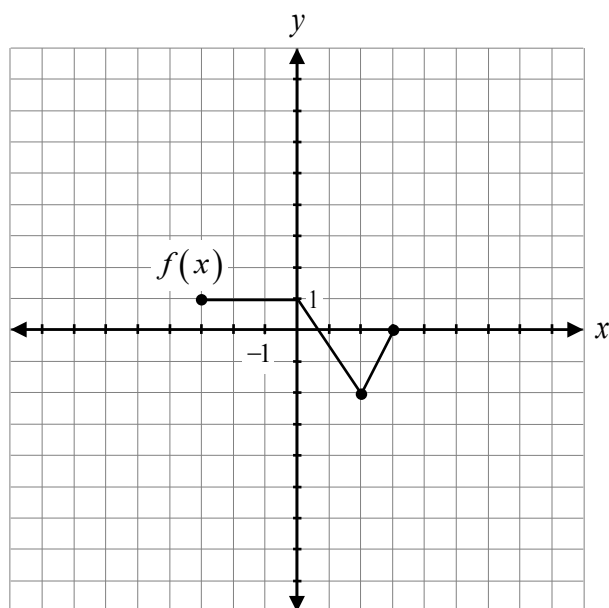
award full marks

– 1 mark for concept error (omitting  $f$ )

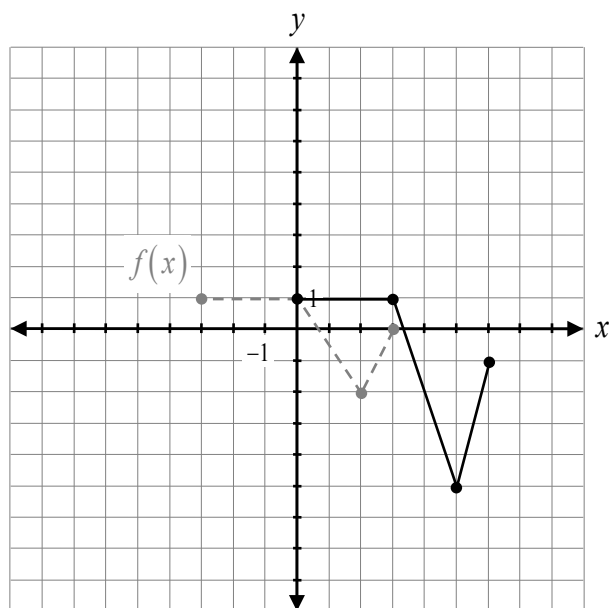
# Question 11

R4

Given the graph of  $f(x)$ , sketch the graph of  $y + 1 = 2f(x - 3)$ .



## Solution

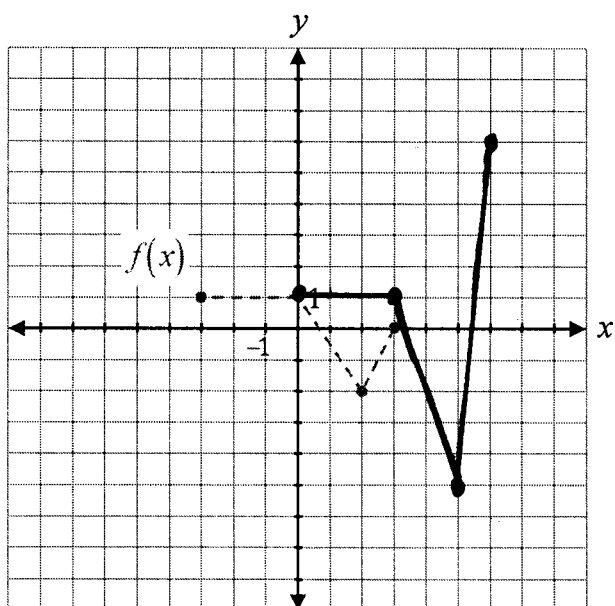


1 mark for vertical stretch  
1 mark for horizontal translation  
1 mark for vertical translation

**3 marks**

## Exemplar 1

---



---

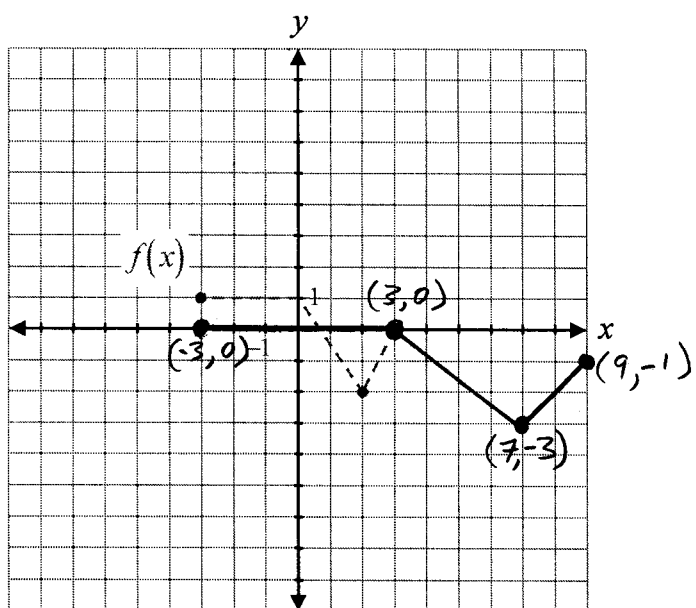
**2½ out of 3**

award full marks

– ½ mark for procedural error (one incorrect point)

## Exemplar 2

---



---

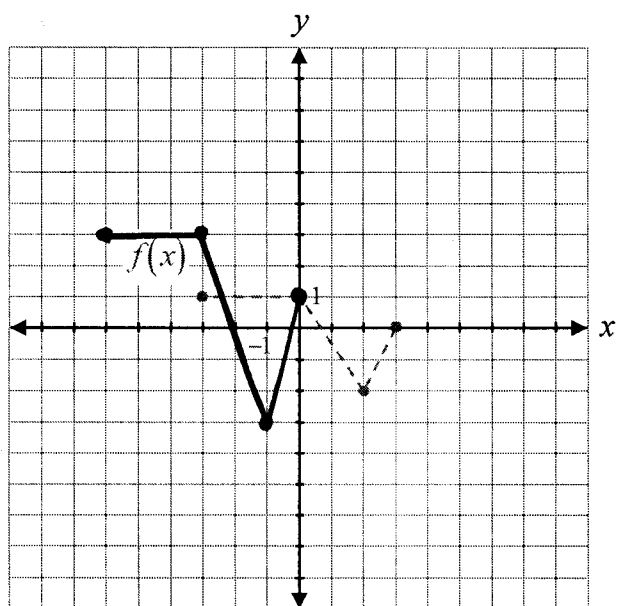
**2 out of 3**

+ 1 mark for horizontal translation

+ 1 mark for vertical translation

## Exemplar 3

---



---

**1 out of 3**

+ 1 mark for vertical stretch

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State the equation of the horizontal asymptote of  $f(x) = \frac{2x^2 - 3x + 5}{4x^2 + 2x - 7}$ .

**Solution**

$$y = \frac{1}{2}$$

**1 mark**

### Exemplar 1

---

$$\begin{aligned} &= \frac{2}{4} \\ \text{H.A.} &= \frac{1}{2} \end{aligned}$$

---

**½ out of 1**

award full marks

– ½ mark for procedural error

### Exemplar 2

---

$$\begin{aligned} \frac{2}{4} &= 2 \\ y &= 2 \end{aligned}$$

---

**½ out of 1**

award full marks

– ½ mark for arithmetic error

### Exemplar 3

---

$$y = 2$$

---

**0 out of 1**

### Exemplar 4

---

Horizontal Asymptote  $x = \frac{1}{2}$

---

**0 out of 1**

### Question 13

R11

Given that  $(x + 4)$  is one of the factors of  $p(x) = x^3 + 6x^2 - 32$ , express  $p(x)$  in completely factored form.

#### Solution

$$\begin{array}{r|rrrrr} -4 & 1 & 6 & 0 & -32 & \\ & \downarrow & -4 & -8 & 32 & \\ \hline & 1 & 2 & -8 & 0 & \end{array}$$

½ mark for  $x = -4$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x + 4)(x^2 + 2x - 8)$$

$$p(x) = (x + 4)(x + 4)(x - 2)$$

or

½ mark for consistent product of factors

**2 marks**

$$p(x) = (x + 4)^2(x - 2)$$

## Exemplar 1

---

$$\begin{array}{r}
 x^2 + 2x - 8 \\
 x+4 \overline{) x^3 + 6x^2 + 0x - 32} \\
 \underline{x^3 + 4x^2} \phantom{+ 0x - 32} \\
 2x^2 - 0x \phantom{- 32} \\
 \underline{2x^2 + 8x} \phantom{- 32} \\
 -8x - 32 \\
 \underline{-8x - 32} \\
 0
 \end{array}$$

$$\begin{aligned}
 &x^2 + 2x - 8 \\
 &(x+4)(x-2)
 \end{aligned}$$

$$p(x) = \underline{(x+4)(x-2)}$$

**1½ out of 2**

+ ½ mark for  $x = -4$

+ 1 mark for equivalent strategy

## Exemplar 2

---

$$\begin{array}{r|rrrr}
 -4 & 1 & 6 & 0 & -32 \\
 & & -4 & -8 & 32 \\
 \hline
 & 1 & 2 & -8 & 0
 \end{array}$$

$$p(x) = \underline{(x^2 + 2x - 8)(x+4)}$$

**1½ out of 2**

+ ½ mark for  $x = -4$

+ 1 mark for synthetic division

### Exemplar 3

---

$$\begin{array}{r|rrrr} -4 & 1 & 6 & 0 & -32 \\ & \downarrow & -4 & -8 & 32 \\ \hline & 1 & 2 & -8 & 0 \end{array}$$

$$x^2 + 2x - 8 = 0$$

$$(x+4)(x-2) = 0$$

$$x = -4 \quad x = 2$$

$$p(x) = \underline{\quad x = -4 \quad x = 2 \quad}$$

---

**1½ out of 2**

+ ½ mark for  $x = -4$

+ 1 mark for synthetic division

### Exemplar 4

---

$$\begin{array}{r|rrr} -4 & 1 & 6 & -32 \\ & \downarrow & -4 & -8 \\ \hline & 1 & 2 & -40 \end{array}$$

$$P(x) = (x^2 + 2x - 40)(x+4)$$

$$p(x) = \underline{(x^2 + 2x - 40)(x+4)}$$

---

**1 out of 2**

+ ½ mark for  $x = -4$

+ 1 mark for synthetic division

– ½ mark for procedural error in line 1

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Prove the identity for all permissible values of  $\theta$ .

$$\frac{2 \cos^2 \theta}{1 - \cot \theta} = \frac{\sin 2\theta}{\tan \theta - 1}$$

### Solution

#### Method 1

| Left-Hand Side                                                          | Right-Hand Side                                                                   |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $\frac{2 \cos^2 \theta}{1 - \frac{\cos \theta}{\sin \theta}}$           | $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta}{\cos \theta} - 1}$           |
| $\frac{2 \cos^2 \theta}{\frac{\sin \theta - \cos \theta}{\sin \theta}}$ | $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta - \cos \theta}{\cos \theta}}$ |
| $\frac{2 \cos^2 \theta \sin \theta}{\sin \theta - \cos \theta}$         | $\frac{2 \sin \theta \cos^2 \theta}{\sin \theta - \cos \theta}$                   |

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

**3 marks**

**Solution****Method 2**

| Left-Hand Side                            | Right-Hand Side                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{2 \cos^2 \theta}{1 - \cot \theta}$ | $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta}{\cos \theta} - 1}$ $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta - \cos \theta}{\cos \theta}}$ $\frac{2 \sin \theta \cos^2 \theta}{\sin \theta - \cos \theta}$ $\frac{\sin \theta (2 \cos^2 \theta)}{\sin \theta \left( 1 - \frac{\cos \theta}{\sin \theta} \right)}$ $\frac{2 \cos^2 \theta}{1 - \cot \theta}$ |

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

**3 marks**

**Solution****Method 3**

| Left-Hand Side                                                                           | Right-Hand Side                        |
|------------------------------------------------------------------------------------------|----------------------------------------|
| $\frac{2 \cos^2 \theta}{1 - \cot \theta}$                                                | $\frac{\sin 2\theta}{\tan \theta - 1}$ |
| $\frac{2 \cos^2 \theta}{1 - \frac{1}{\tan \theta}}$                                      |                                        |
| $\frac{2 \cos^2 \theta}{\frac{\tan \theta - 1}{\tan \theta}}$                            |                                        |
| $\frac{2 \cos^2 \theta}{\tan \theta - 1} (\tan \theta)$                                  |                                        |
| $\frac{2 \cos^2 \theta \left( \frac{\sin \theta}{\cos \theta} \right)}{\tan \theta - 1}$ |                                        |
| $\frac{2 \cos \theta \sin \theta}{\tan \theta - 1}$                                      |                                        |
| $\frac{\sin 2\theta}{\tan \theta - 1}$                                                   |                                        |

1 mark for algebraic strategies

1 mark for logical process to prove the identity

1 mark for correct substitution of identities

**3 marks**

## Exemplar 1

---

| Left-Hand Side                                                                                  | Right-Hand Side                                                                                  |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $\rightarrow \frac{2\cos^2\theta}{1-\cot\theta}$                                                | $\rightarrow \frac{\sin 2\theta}{\tan\theta - 1}$                                                |
| $\rightarrow \frac{2\cos^2\theta}{\frac{\sin\theta}{\sin\theta} \frac{\cos\theta}{\sin\theta}}$ | $\rightarrow \frac{\sin 2\theta}{\frac{\sin\theta}{\cos\theta} - \frac{\cos\theta}{\cos\theta}}$ |
| $\rightarrow 2\cos^2\theta \left( \frac{\sin\theta}{\sin\theta\cos\theta} \right)$              | $\rightarrow \frac{\sin 2\theta}{\frac{\sin\theta \cdot \cos\theta}{\cos\theta}}$                |
| $\rightarrow \frac{2\cos^2\theta \sin\theta}{\sin\theta\cos\theta}$                             | $\rightarrow \frac{2\sin\theta\cos\theta(\cos\theta)}{\sin\theta\cos\theta}$                     |
| $\rightarrow \frac{2\cos\theta\sin\theta}{\sin\theta}$                                          |                                                                                                  |
| $\rightarrow 2\cos\theta$                                                                       | $\rightarrow 2\cos\theta$                                                                        |

$\therefore \text{LHS} = \text{RHS}$

---

**2 out of 3**

- + 1 mark for correct substitution of identities
- + 1 mark for logical process to prove the identity

## Exemplar 2

| Left-Hand Side                                                                      | Right-Hand Side                                                                               |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| $\frac{2\cos^2\theta}{1-\cot\theta}$                                                | $\frac{\sin 2\theta}{\tan\theta - 1}$                                                         |
| $\frac{2\cos^2\theta}{\frac{\sin\theta}{\sin\theta} \frac{\cos\theta}{\sin\theta}}$ | $\frac{2\sin\theta\cos\theta}{\tan\theta - 1}$                                                |
|                                                                                     | $\frac{2\sin\theta\cos\theta}{\frac{\sin\theta}{\cos\theta} - \frac{\cos\theta}{\cos\theta}}$ |
|                                                                                     | $\frac{2\sin\theta\cos\theta}{\frac{\sin\theta - \cos\theta}{\cos\theta}}$                    |

**2 out of 3**

- + 1 mark for correct substitution of identities
- + 1 mark for algebraic strategies

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Restaurant A has 5 types of hamburgers, 2 types of french fries, and 10 types of drinks.

Restaurant B has 4 types of hamburgers, 5 types of french fries, and 6 types of drinks.

If a meal is made up of a hamburger, french fries, and a drink, justify which restaurant offers a greater variety of meals.

**Solution**

Restaurant A:  $5 \cdot 2 \cdot 10 = 100$  meals

Restaurant B:  $4 \cdot 5 \cdot 6 = 120$  meals

Restaurant B offers a greater variety of meals.

**1 mark**

### Exemplar 1

---

$$A (5)(2)(10) = 1000$$

$$B (5)(6)(4) = 120$$

A has more variety.

---

**½ out of 1**

award full marks

– ½ mark for arithmetic error in line 1

### Exemplar 2

---

Restaurant B

---

**0 out of 1**

Express  $\log_7(2x - 5) + 2\log_7 3$  as a single logarithm.

**Solution**

$$\log_7(2x - 5) + \log_7 3^2 \quad 1 \text{ mark for power law}$$

$$\log_7(2x - 5) + \log_7 9$$

$$\log_7(9(2x - 5)) \quad 1 \text{ mark for product law}$$

**or**

**2 marks**

$$\log_7(18x - 45)$$

### Exemplar 1

---

$$\log_7 (2x-5) + \log_7 3^2$$

$$\log_7 (2x-5)(9)$$

$$\log_7 18x-45$$

---

**2 out of 2**

award full marks

E7 (notation error in lines 2 and 3)

### Exemplar 2

---

$$\log_7 (2x-5) + \log_7 3^2$$

$$\log_7 ((2x-5)(6))$$

$$\log (12x-30)$$

---

**1½ out of 2**

award full marks

– ½ mark for arithmetic error in line 2

E7 (transcription error in line 3)

### Exemplar 3

---

$$\log_7 2x - \log_7 5 + \log_7 9$$

$$\log_7 \left( \frac{(2x)(9)}{5} \right)$$

---

**1 out of 2**

award full marks

– 1 mark for concept error

---

Explain why  $11!$  is not the total number of 11-letter arrangements that can be made from the word CELEBRATION.

**Solution**

The total number of possible arrangements is half of  $11!$  because switching the two Es creates duplicate arrangements.

**1 mark**

### Exemplar 1

---

because celebration has 2 E's so you have  
to consider that!

$$\frac{11!}{2!} = 19958400$$

---

**½ out of 1**

award full marks

– ½ mark for lack of clarity in explanation

### Exemplar 2

---

there are duplicate E's

---

**½ out of 1**

award full marks

– ½ mark for lack of clarity in explanation

### Exemplar 3

---

Because the letter E repeats  
we subtract 2!

---

**0 out of 1**

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## Scoring Guidelines for Booklet 2 Questions

## Answer Key for Selected Response Questions

| Question | Answer | Learning Outcome |
|----------|--------|------------------|
| 18       | C      | R13              |
| 19       | B      | P2               |
| 20       | D      | T3               |
| 21       | D      | R7               |
| 22       | C      | T4               |
| 23       | A      | R12              |
| 24       | B      | P4               |
| 25       | A      | R2               |
| 26       | B      | R11              |

### Question 18

R13

Given  $f(x) = x - 1$ , identify a point on the graph of  $y = \sqrt{f(x)}$ .

a)  $(0, -1)$

b)  $(3, 2)$

c)  $(1, 0)$

d)  $(0, 1)$

### Question 19

P2

Identify the total number of possible arrangements of 6 adults and 4 children seated in a row if the children must sit together.

a)  $6!4!$

b)  $7!4!$

c)  $10!$

d)  $6!$

### Question 20

T3

Identify the exact value of  $\sec\left(-\frac{7\pi}{3}\right)$ .

a)  $-2$

b)  $-\frac{2}{\sqrt{3}}$

c)  $\frac{2}{\sqrt{3}}$

d)  $2$

### Question 21

R7

Given  $\log_x \left( \frac{1}{25} \right) = -2$ , identify the value of  $x$ .

a)  $-5$

b)  $-\frac{1}{5}$

c)  $\frac{1}{5}$

d)  $5$

### Question 22

T4

Identify the equation for all of the asymptotes on the graph of  $y = \tan x$ .

a)  $x = k\pi, k \in \mathbb{Z}$

b)  $x = 2k\pi, k \in \mathbb{Z}$

c)  $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

d)  $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

### Question 23

R12

If  $p(x) = 3(m)(x+1)^2$  is a cubic function with a  $y$ -intercept of  $-12$ , identify the missing factor,  $m$ .

a)  $m = x - 4$

b)  $m = x + 4$

c)  $m = x + 12$

d)  $m = x - 12$

### Question 24

P4

Identify the number of negative terms in the binomial expansion of  $(x - y)^5$ .

a) 2

b) 3

c) 5

d) 6

### Question 25

R2

Given  $f(x) = x^2$ , identify which equation represents the graph of  $y = f(x)$  after a translation of 5 units to the left.

a)  $y = (x + 5)^2$

b)  $y = (x - 5)^2$

c)  $y = x^2 - 5$

d)  $y = x^2 + 5$

### Question 26

R11

When a polynomial,  $p(x)$ , is divided by  $(x - 7)$ , the remainder is 24. Identify the only statement that must be true.

a)  $x = 7$  is a zero of  $p(x)$

b)  $p(7) = 24$

c)  $x = 24$  is a zero of  $p(x)$

d) the y-intercept is 24

---

Given  $f(x) = \frac{(2x+1)(x-8)}{(x-8)(x+4)}$ , state the equation(s) of the vertical asymptote(s).

**Solution**

$$x = -4$$

**1 mark**

### Exemplar 1

---

$$V.A.: y = -4$$

---

**0 out of 1**

### Exemplar 2

---

$$f(x) = \frac{(2x+1)}{x+4}$$

$$x \neq 8, -4$$

---

**0 out of 1**

### Exemplar 3

---

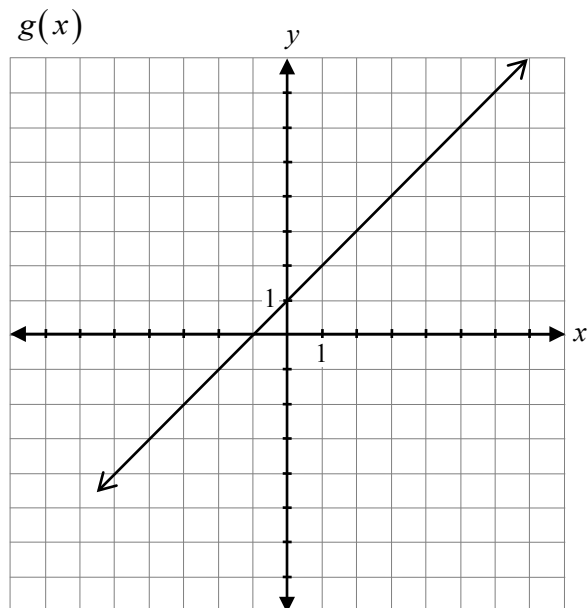
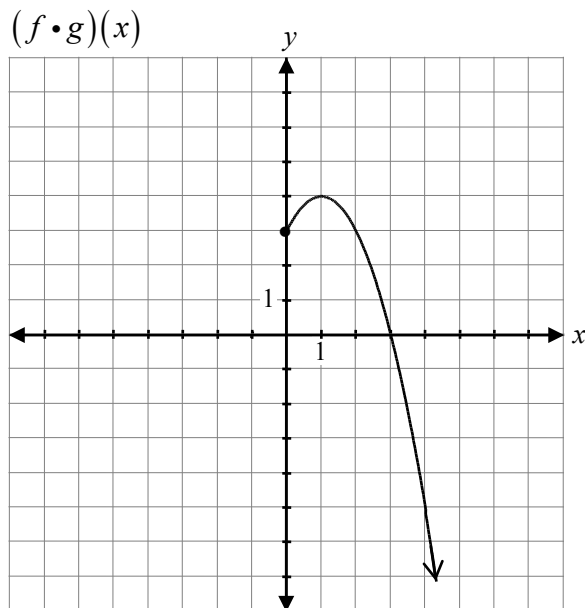
$$x \neq -4$$

---

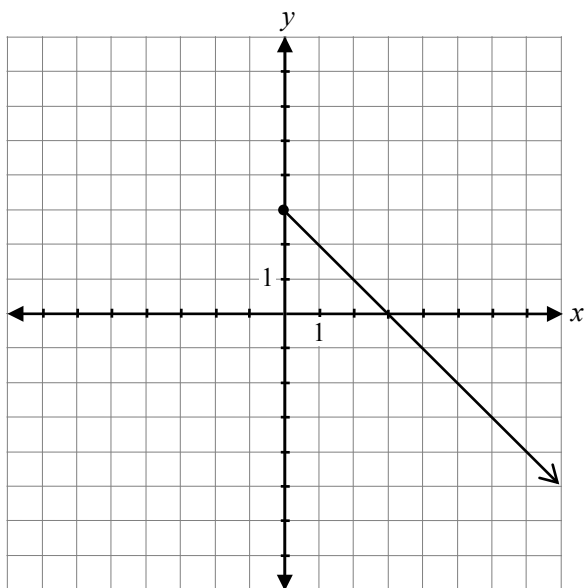
**1 out of 1**

E7 (notation error)

Given the graphs of  $(f \cdot g)(x)$  and  $g(x)$ , sketch the graph of  $f(x)$ .



### Solution

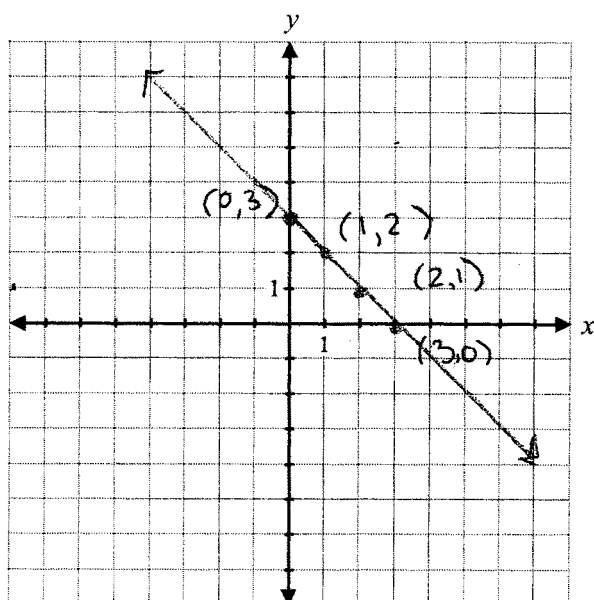


1 mark for operation of division  
1 mark for restricted domain

**2 marks**

## Exemplar 1

---

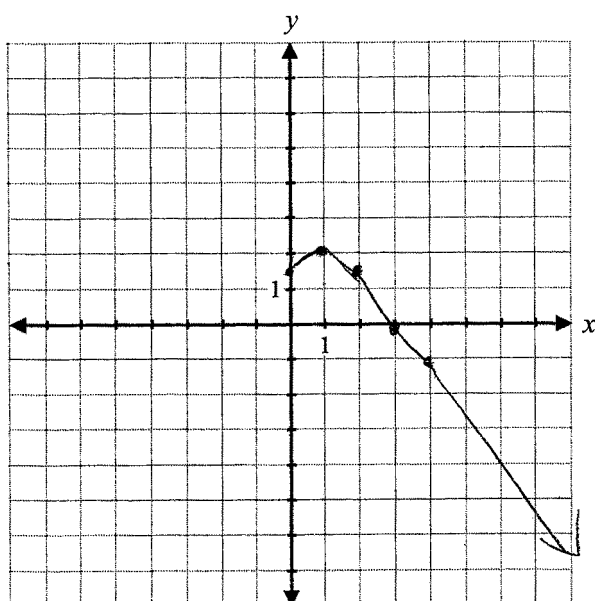


**1 out of 2**

+ 1 mark for operation of division

## Exemplar 2

---

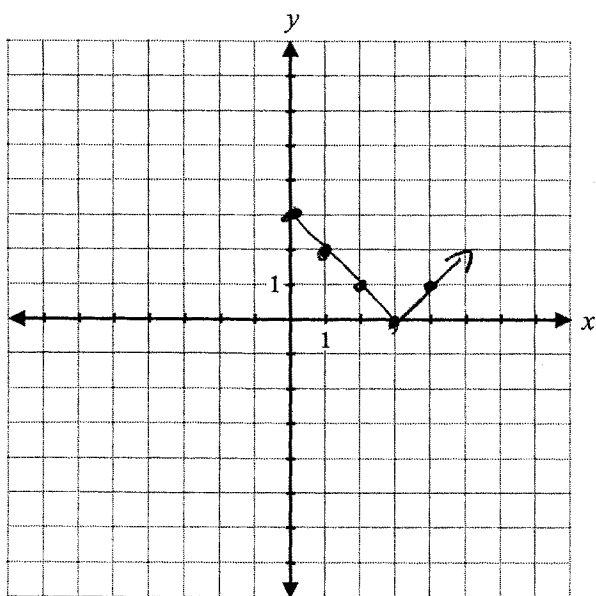


**1 out of 2**

+ 1 mark for restricted domain

### Exemplar 3

---



---

**1½ out of 2**

award full marks

– ½ mark for procedural error (one incorrect point)

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Brian was asked to state the zeros of the polynomial  $p(x) = (x + 2)(x - 5)(x - 1)$ .

Brian's response:

$$\text{Zeros: } (x+2), (x-5), (x-1)$$

Explain why his response is incorrect.

**Solution**

Brian stated the factors instead of the zeros.

**1 mark**

### Exemplar 1

---

The zeros are  $(x+2)(x-5)(x-1)$  after they have been solved:

$$x+2=0$$

$$x-5=0$$

$$x-1=0$$

$$x=-2$$

$$x=5$$

$$x=1$$

---

0 out of 1

### Exemplar 2

---

the 0's are what make the  
inner value of the brackets 0  
 $x=-2, 5, 1$  would be correct

---

1 out of 1

Simplify  ${}_{n+3}C_2$ .

**Solution**

$$\frac{(n+3)!}{(n+3-2)!2!}$$

½ mark for substitution

$$\frac{(n+3)!}{(n+1)!2!}$$

$$\frac{(n+3)(n+2)\cancel{(n+1)!}}{\cancel{(n+1)!}2}$$

1 mark for factorial expansion

$$\frac{(n+3)(n+2)}{2}$$

½ mark for simplification of factorials

or

**2 marks**

$$\frac{n^2 + 5n + 6}{2}$$

## Exemplar 1

---

$$\frac{(n+3)!}{((n+3)-2)! 2!}$$

$$\frac{(n+3)!}{(n+1)! 2}$$

$$\frac{\cancel{(n+3)!}}{(n+1)(n+2)\cancel{(n+3)!} 2}$$

$$\frac{1}{(n+1)(n+2) 2}$$

---

**1 out of 2**

+ ½ mark for substitution

+ ½ mark for simplification of factorials

## Exemplar 2

---

$$\frac{n+3!}{2!((n+3)-2)!}$$

$$\frac{n+3!}{2!(n+1)!}$$

$$\frac{(n+3)(n+2)(\cancel{n+1})!}{2!(\cancel{n+1})!}$$

$$\frac{n^2 + 5n + 6}{2!}$$

---

**1½ out of 2**

+ ½ mark for substitution

+ 1 mark for factorial expansion

E4 (missing brackets but still implied in lines 1 and 2)

### Exemplar 3

---

$$\frac{(n+3)!}{(n+3-2)! 2!}$$

$$\frac{(n+3)!}{(n+1)! 2!}$$

$$\frac{(n+3)(n+2)(\cancel{n+1})!}{(\cancel{n+1})! 2!}$$

$$\frac{(n+3)(n+2)}{2} = 0.2$$

$$n = -3 \quad n = -2$$

---

**1½ out of 2**

award full marks

– ½ mark for procedural error (solving for  $n$ )

E1 (impossible solution not rejected in final answer)

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Verify that the equation  $2 \cos^2 x = \sin x + 1$  is true for  $x = \frac{\pi}{6}$ .

### Solution

| Left-Hand Side                          | Right-Hand Side                         |                                                        |
|-----------------------------------------|-----------------------------------------|--------------------------------------------------------|
| $2 \cos^2 \left( \frac{\pi}{6} \right)$ | $\sin \left( \frac{\pi}{6} \right) + 1$ | $\frac{1}{2}$ mark for substitution                    |
| $2 \left( \frac{\sqrt{3}}{2} \right)^2$ | $\frac{1}{2} + 1$                       | 1 mark for exact values ( $\frac{1}{2}$ mark for each) |
| $2 \left( \frac{3}{4} \right)$          |                                         |                                                        |
| $\frac{3}{2}$                           | $\frac{3}{2}$                           | $\frac{1}{2}$ mark for simplification                  |
| $\therefore \text{LHS} = \text{RHS}$    |                                         | <b>2 marks</b>                                         |

### Exemplar 1

---

$$2 \cos^2 x = \sin x + 1$$

$$2 \cos^2 x = \sin x + 1$$

$$2 \cos^2 x - 1 = \sin x$$

$$\cos 2x = \sin x$$

$$\cos 2\left(\frac{\pi}{6}\right) = \sin \frac{\pi}{6}$$

$$\cos \frac{\pi}{3} = \sin \frac{\pi}{6}$$

$$\frac{1}{2} = \frac{1}{2} \quad \checkmark$$

---

**2 out of 2**

### Exemplar 2

---

$2 \cos^2 x = \sin x + 1$  is  
not true for  $x = \pi/6$

$$2 \cos^2 x = \sin x + 1$$

$$2 \cos^2 (\pi/6) = \sin(\pi/6) + 1$$

$$2(\sqrt{3}/2)^2 = (1/2) + 1$$

$$2(3/4) = 3/2$$

$$6/4 \neq 3/2$$

---

**1½ out of 2**

award full marks

– ½ mark for arithmetic error in line 5

### Exemplar 3

---

$$2\cos^2 x = \sin x + 1$$

$$2\cos^2 x - \sin x - 1 = 0$$

$$2(1 - \sin^2 x) - \sin x - 1 = 0$$

$$2 - 2\sin^2 x - \sin x - 1 = 0$$

$$\begin{array}{r} -2\sin^2 x - \sin x + 1 = 0 \\ \hline -1 \quad -1 \quad -1 \quad -1 \end{array}$$

$$2\sin^2 x + \sin x - 1 = 0$$

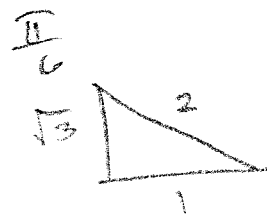
$$(2\sin x - 1)(\sin x + 1) = 0$$

$$2\sin x - 1 = 0$$

$$\sin x = \frac{1}{2}$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}$$

yes, it is true.



---

2 out of 2

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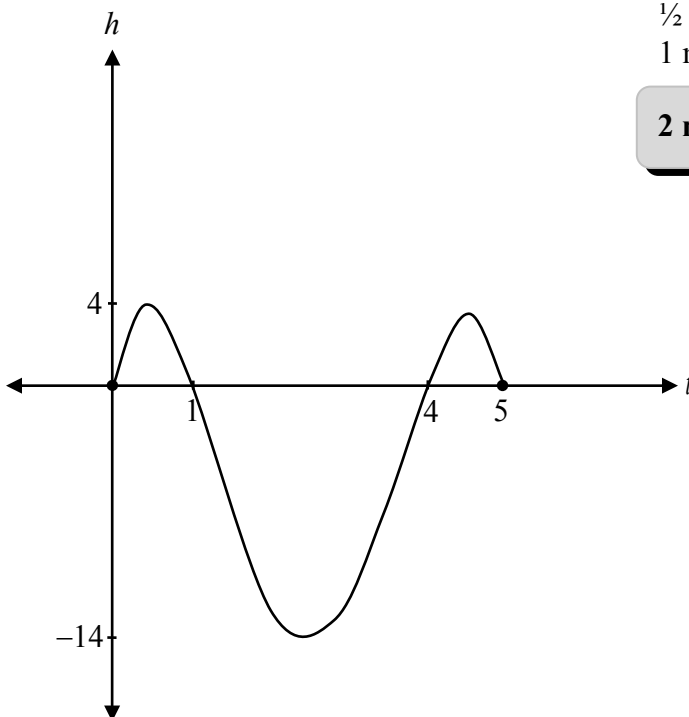
The height of a fish jumping out of the water can be modelled by the function

$h(t) = -t(t-1)(t-4)(t-5)$  where  $h(t)$  is the height of the fish above or below the water in cm, and  $t$  is the time in seconds,  $t \geq 0$ .

- a) Sketch a graph representing the height of the fish with respect to time over the interval  $[0, 5]$ .
- b) State, from the graph in a), the total number of seconds that the fish is above the water.

**Solution**

a)



½ mark for  $h$ -intercept

½ mark for other  $t$ -intercepts

1 mark for vertical reflection of a quartic function

**2 marks**

b) 2 seconds

1 mark for time consistent with graph in a)

**1 mark**

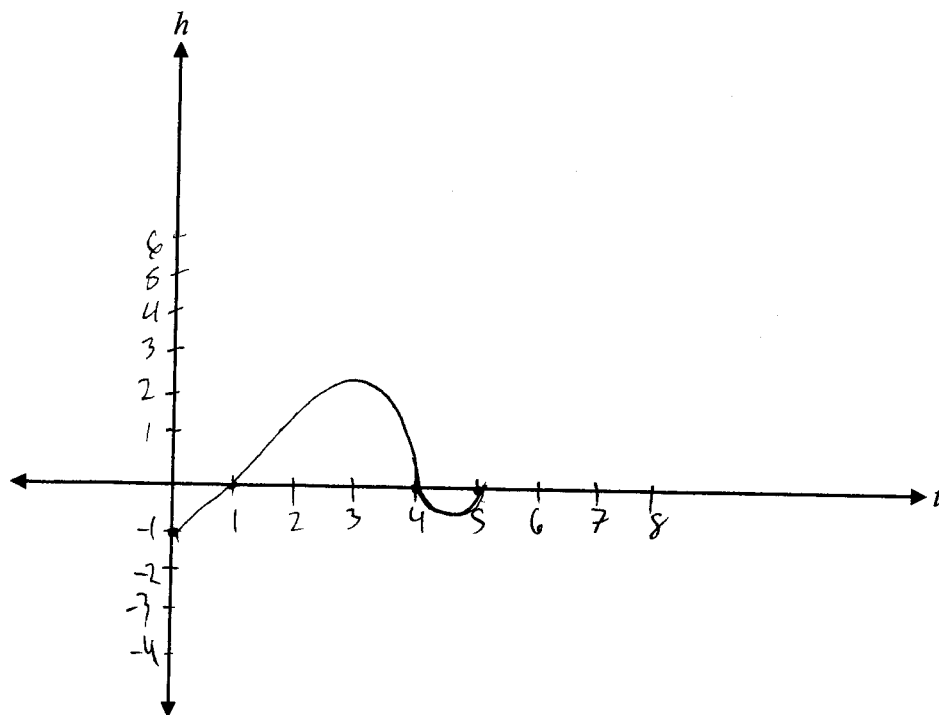
Note:

- Scale values on  $h$ -axis are not required.

## Exemplar 1

---

a)



---

**½ out of 2**

+ ½ mark for other  $t$ -intercepts

---

b)

2 seconds

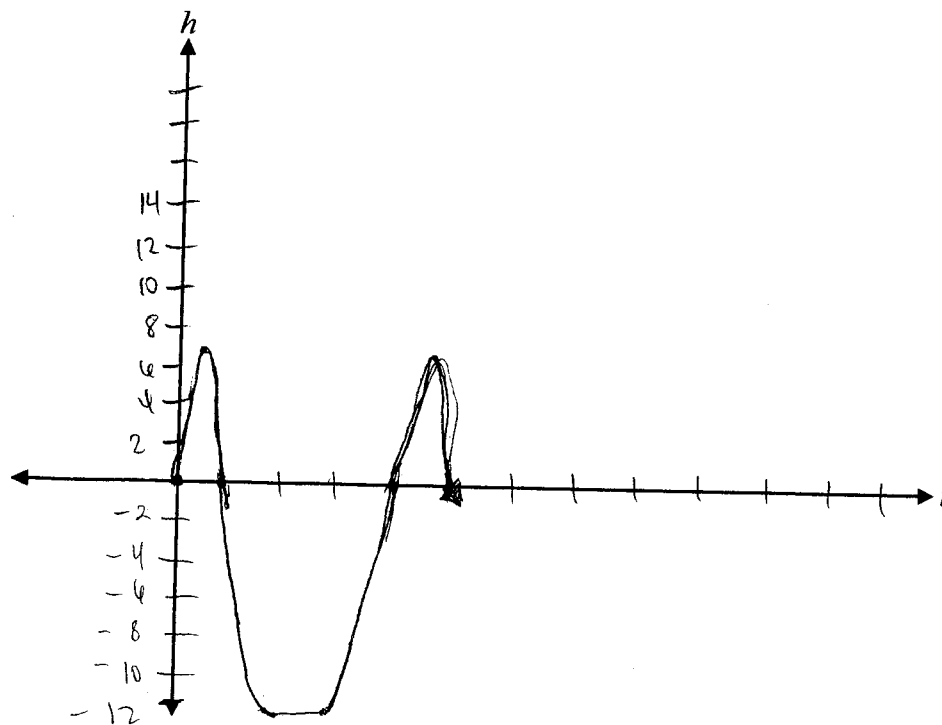
---

**0 out of 1**

## Exemplar 2

---

a)



---

**2 out of 2**

award full marks

E9 (scale values on  $t$ -axis not indicated)

E8 (answer outside the given domain)

---

b)

2 seconds.

---

**1 out of 1**

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Describe the behaviour of the graph of  $y = 5^x + 4$  as it approaches  $y = 4$ .

**Solution**

There is a horizontal asymptote at  $y = 4$ , so the graph approaches  $y = 4$  without ever touching it.

**1 mark**

### Exemplar 1

---

- The graph will not pass through  $y=4$   
Since it is an asymptote

---

**1 out of 1**

### Exemplar 2

---

The  $y$  value will come close to 4 but won't actually reach 4. it will grow smaller and smaller but it will never reach 0

---

**½ out of 1**

award full marks

– ½ mark for lack of clarity in description

### Exemplar 3

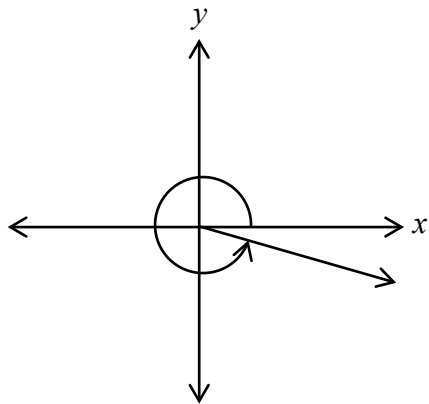
---

When it approaches  $y=4$ , the graph will get very close to the vertical asymptote there, and as it gets closer, the  $x$  values will increase rapidly.

---

**0 out of 1**

Sketch the angle of 6 radians in standard position.

**Solution**

$\frac{1}{2}$  mark for an angle in quadrant IV  
 $\frac{1}{2}$  mark for an appropriate angle

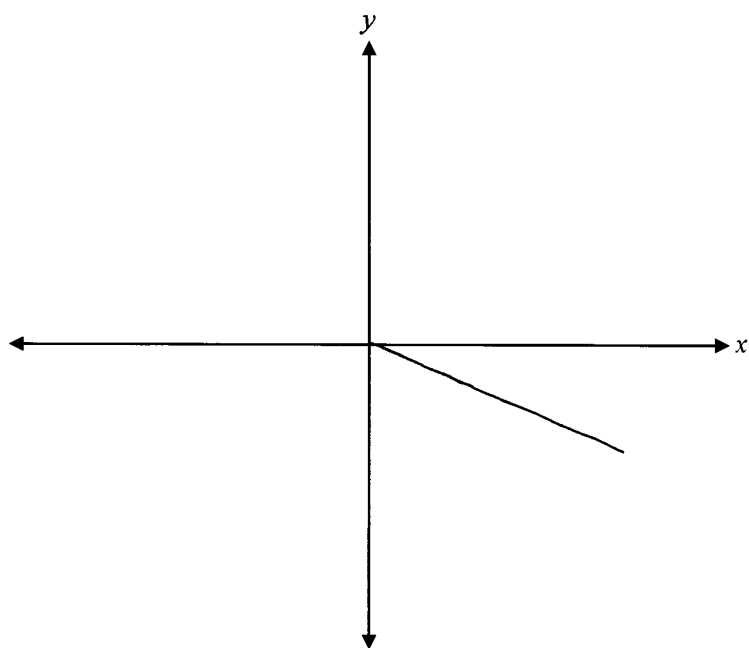
**1 mark**

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated).

## Exemplar 1

---



---

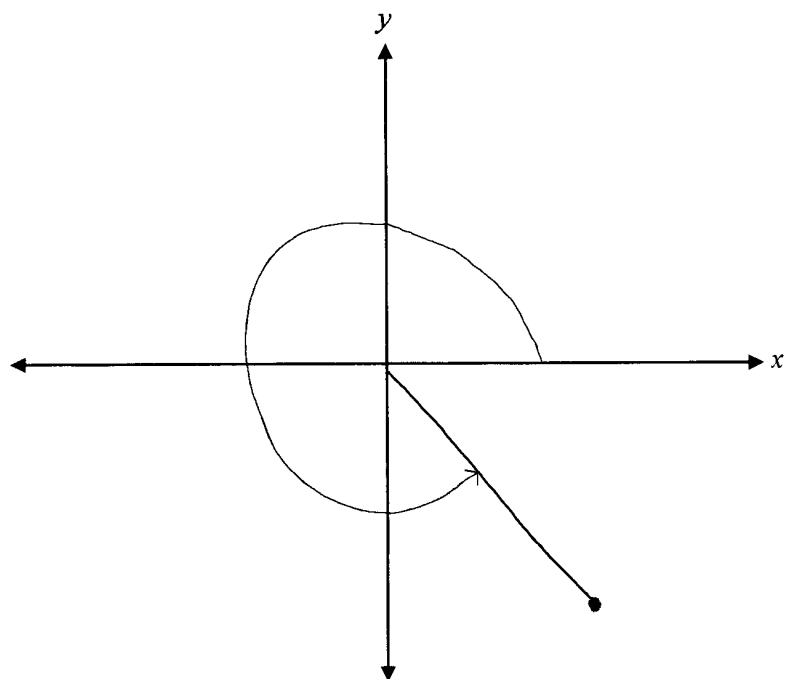
**1 out of 1**

award full marks

E1 (final answer not stated)

## Exemplar 2

---

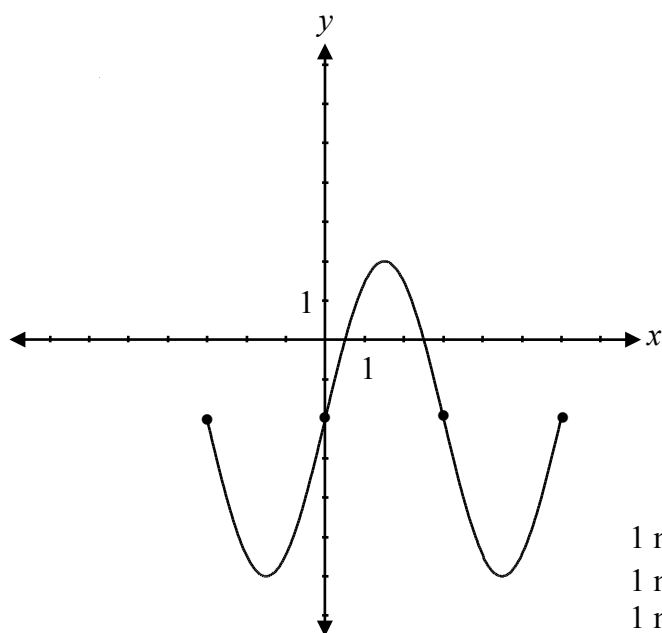


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**½ out of 1**

+ ½ mark for an angle in quadrant IV

Sketch the graph of the function  $y = 4 \sin\left(\frac{\pi}{3}x\right) - 2$  over the domain  $[-3, 6]$ .

**Solution**

1 mark for shape of  $y = \sin x$   
1 mark for amplitude  
1 mark for period  
1 mark for vertical translation

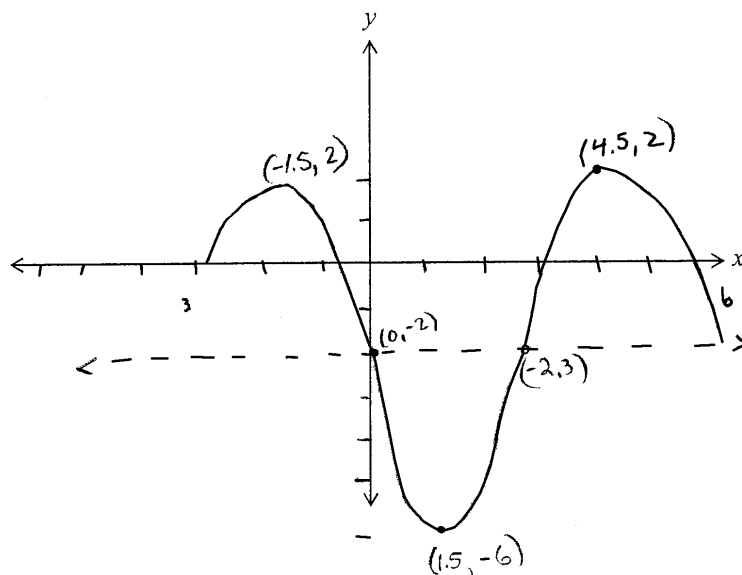
**4 marks**

Note:

- Deduct  $\frac{1}{2}$  mark for procedural error for not completing the domain of  $[-3, 6]$ .
- If period mark not awarded, do not deduct for E9 error (scale value on  $x$ -axis not indicated).

## Exemplar 1

---

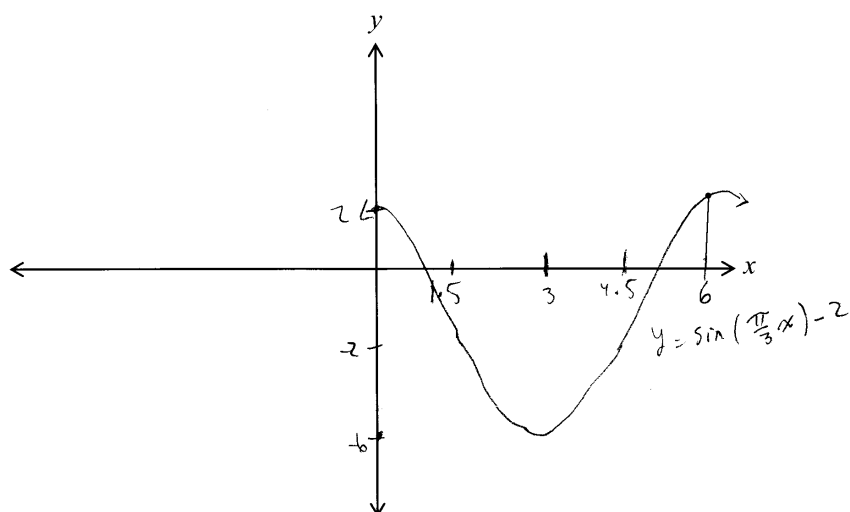


**2½ out of 4**

- + 1 mark for amplitude
- + 1 mark for period
- + 1 mark for vertical translation
- ½ mark for procedural error (not completing the domain of  $[-3, 6]$ )

## Exemplar 2

---

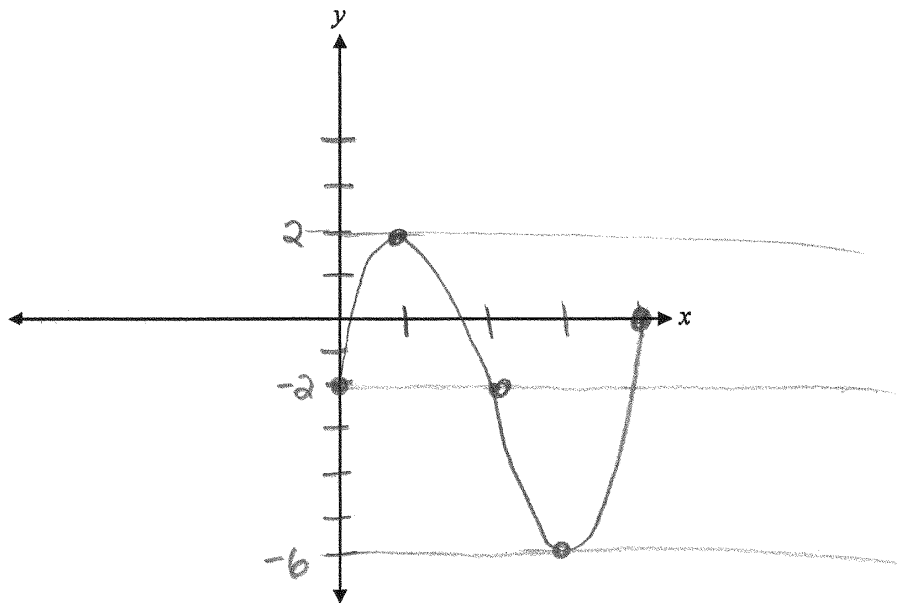


**2½ out of 4**

- + 1 mark for amplitude
- + 1 mark for period
- + 1 mark for vertical translation
- ½ mark for procedural error (not completing the domain of  $[-3, 6]$ )

### Exemplar 3

---



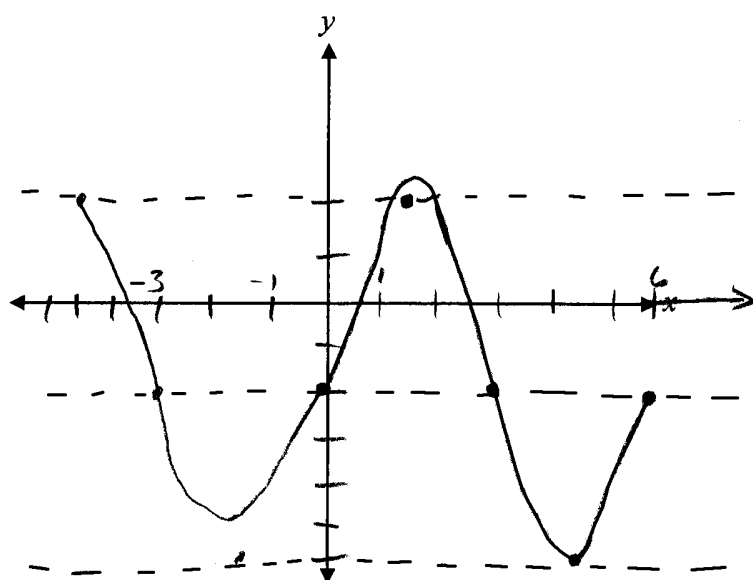
---

**2 out of 4**

- + 1 mark for shape of  $y = \sin x$
- + 1 mark for amplitude
- + 1 mark for vertical translation
- ½ mark for procedural error (one incorrect point)
- ½ mark for procedural error (not completing the domain of  $[-3, 6]$ )

## Exemplar 4

---



---

**3 out of 4**

+ 1 mark for amplitude

+ 1 mark for period

+ 1 mark for vertical translation

E8 (answer outside the given domain)

E9 (scale values on y-axis not indicated)

## Question 36

R1, R14

Given  $f(x) = 3x - 12$  and  $g(x) = x - 4$ ,

a) determine the equation of  $h(x) = \left(\frac{f}{g}\right)(x)$ .

b) describe what the non-permissible value represents on the graph of  $h(x)$ .

### Solution

$$\text{a) } h(x) = \frac{3x - 12}{x - 4}$$

1 mark for division of  $\left(\frac{f}{g}\right)(x)$

$$h(x) = \frac{3(\cancel{x-4})}{(\cancel{x-4})}$$

1 mark

$$h(x) = \underline{3, x \neq 4}$$

b) The non-permissible value is a point of discontinuity (hole).

1 mark

### Exemplar 1

---

a) 
$$h(x) = \frac{3x - 12}{x - 4}$$
$$h(x) = \frac{3(\cancel{x-4})}{\cancel{x-4}}$$
$$h(x) = 3$$

---

**1 out of 1**

---

b)

The point at  $x = 4$  doesn't exist.

---

**½ out of 1**

award full marks

– ½ mark for lack of clarity in description

### Exemplar 2

---

a)

$$h(x) = \frac{3x - 12}{x - 4}$$

---

**1 out of 1**

---

b)

$$x \neq 4$$

---

**0 out of 1**

---

Determine an equation of a radical function,  $f(x)$ , with a domain of  $x \geq 5$  and a range of  $y \geq -2$ .

**Solution**

$$f(x) = \sqrt{x-5} - 2$$

1 mark for horizontal translation

1 mark for vertical translation

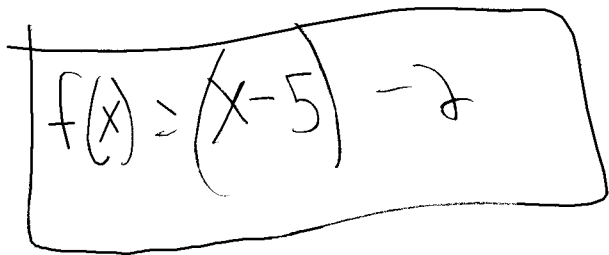
**2 marks**

Note:

- Other equations are possible.

## Exemplar 1

---


$$f(x) = (x-5) - 2$$

---

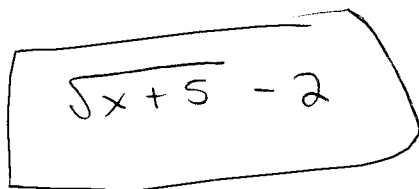
**1 out of 2**

award full marks

– 1 mark for concept error (not writing  $f(x)$  as a radical function)

## Exemplar 2

---


$$\sqrt{x+5} - 2$$

$$f(x) = \underline{\hspace{2cm}}$$

---

**1 out of 2**

+ 1 mark for vertical translation

E2 (changing an equation to an expression)

Determine the exact value of  $\cos\left(\frac{17\pi}{12}\right)$ .

**Solution**

$$\begin{aligned}\cos\left(\frac{17\pi}{12}\right) &= \cos\left(\frac{3\pi}{4} + \frac{2\pi}{3}\right) \\ &= \cos\frac{3\pi}{4}\cos\frac{2\pi}{3} - \sin\frac{3\pi}{4}\sin\frac{2\pi}{3}\end{aligned}$$

1 mark for substitution into correct identity

$$\begin{aligned}&= \left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

2 marks for exact values ( $\frac{1}{2}$  mark for each)

**3 marks**

Note:

- Other combinations are possible.
- Deduct a maximum  $\frac{1}{2}$  mark for arithmetic errors in simplification.

## Exemplar 1

---

$$\begin{aligned} & \cos\left(\frac{8\pi}{12} + \frac{9\pi}{12}\right) \\ & \cos\left(\overset{a}{\frac{2\pi}{3}} + \overset{b}{\frac{3\pi}{4}}\right) \\ & (\cos a \cos b) - (\sin a \sin b) \\ & \left(\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}\right) - \left(-\frac{1}{2} \cdot \frac{\sqrt{2}}{2}\right) \\ & \frac{\sqrt{6}}{2} - \left(-\frac{\sqrt{2}}{2}\right) \end{aligned}$$

---

**1 out of 3**

+ 1 mark for substitution into correct identity

+ ½ mark for the value of  $\sin \frac{3\pi}{4}$

– ½ mark for arithmetic error in line 5

## Exemplar 2

---

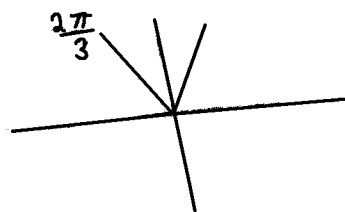
$$\frac{8\pi}{12} + \frac{9\pi}{12} = \frac{17\pi}{12}$$

$$\frac{2\pi}{3} + \frac{3\pi}{4}$$

$$\cos\left(\frac{2\pi}{3}\right) + \cos\left(\frac{3\pi}{4}\right)$$
$$= -\frac{1}{2}(\sqrt{2}) - \frac{1}{\sqrt{2}}(2)$$

$$= \frac{-\sqrt{2}}{2\sqrt{2}} + \frac{-2}{2\sqrt{2}}$$

$$= \frac{-2 - \sqrt{2}}{2\sqrt{2}}$$



---

**1 out of 3**

+ 1 mark for values of  $\cos\frac{2\pi}{3}$  and  $\cos\frac{3\pi}{4}$

### Exemplar 3

---

$$\begin{aligned}\cos \frac{17\pi}{12} &= \cos \frac{3\pi}{4} \cos \frac{2\pi}{3} - \sin \frac{3\pi}{4} \sin \frac{2\pi}{3} \\ &\cos\left(-\frac{\sqrt{2}}{2}\right) \cos\left(-\frac{1}{2}\right) - \sin\left(\frac{\sqrt{2}}{2}\right) \sin\left(\frac{\sqrt{3}}{2}\right) \\ &\left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)\end{aligned}$$

---

**2½ out of 3**

award full marks

– ½ mark for procedural error in line 2

E2 (changing an equation to an expression in lines 2 and 3)

E1 (final answer not stated)

### Exemplar 4

---

$$\begin{aligned}\cos \frac{3\pi}{4} \cos \frac{2\pi}{3} - \sin \frac{3\pi}{4} \sin \frac{2\pi}{3} \\ \cos\left(-\frac{\sqrt{2}}{2}\right) \cos\left(-\frac{1}{2}\right) - \sin\left(\frac{\sqrt{2}}{2}\right) \sin\left(\frac{\sqrt{3}}{2}\right)\end{aligned}$$

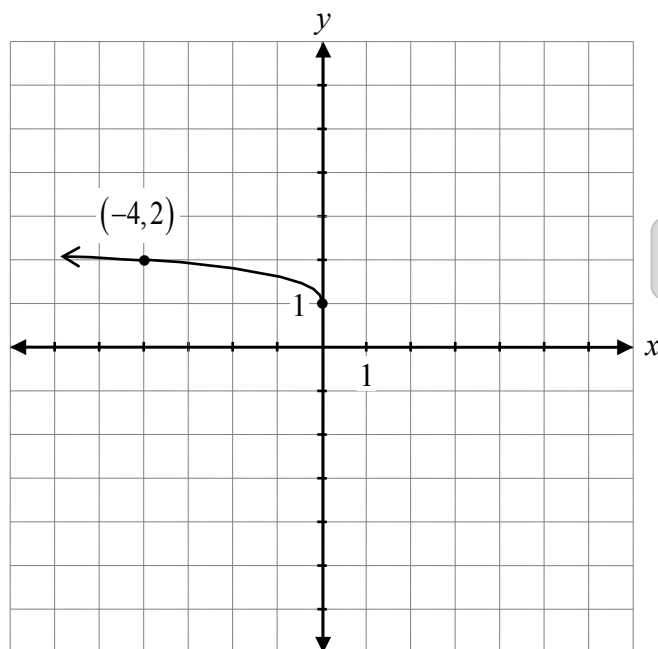
---

**2 out of 3**

award full marks

– 1 mark for concept error in line 2

Sketch the graph of  $y = \frac{1}{2}\sqrt{-x} + 1$ .

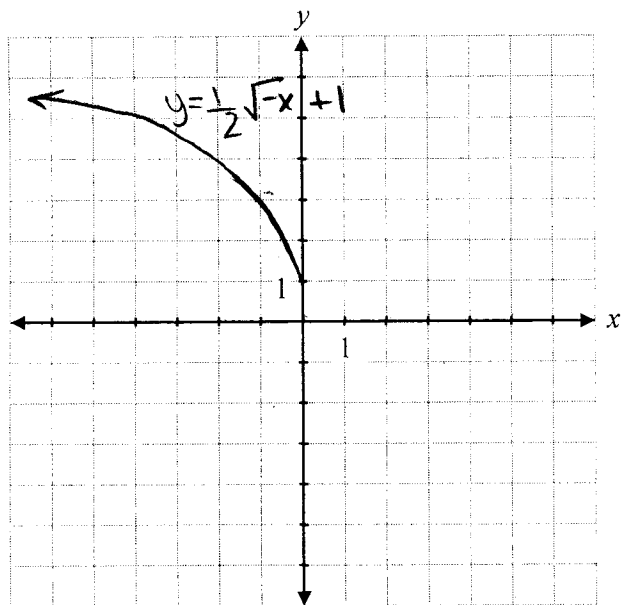
**Solution**

1 mark for shape of a radical function  
1 mark for vertical compression  
1 mark for horizontal reflection  
1 mark for vertical translation

**4 marks**

## Exemplar 1

---

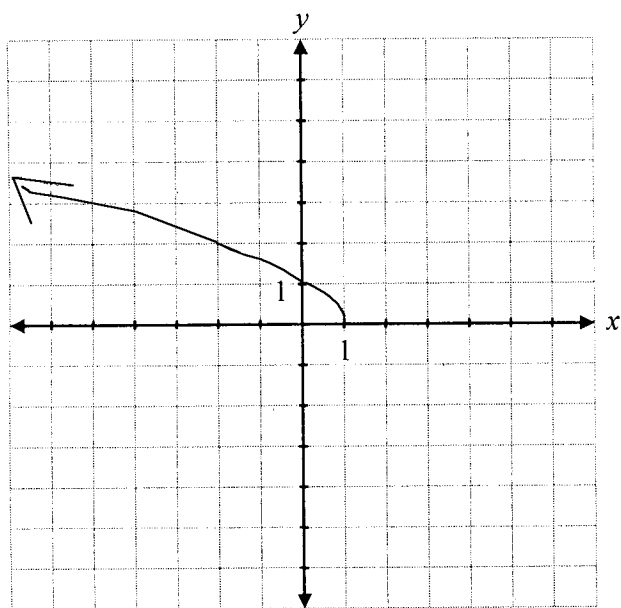


**3 out of 4**

- + 1 mark for shape of a radical function
- + 1 mark for horizontal reflection
- + 1 mark for vertical translation

## Exemplar 2

---

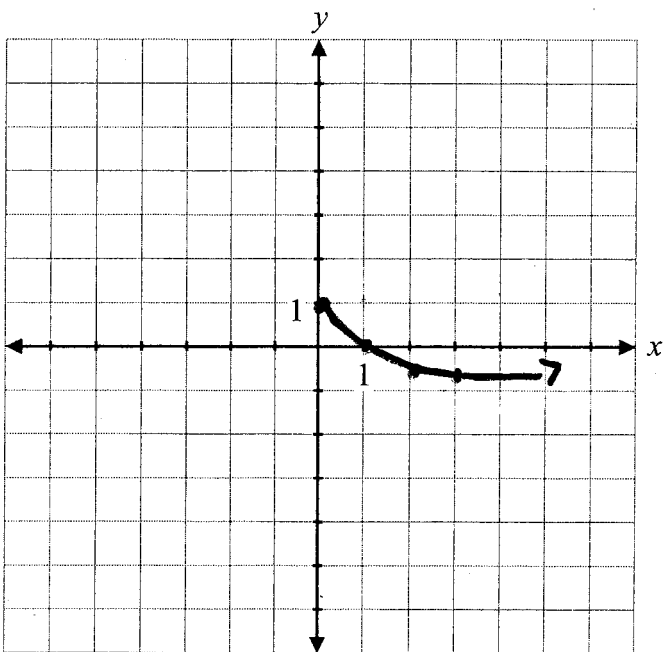


**2 out of 4**

- + 1 mark for shape of a radical function
- + 1 mark for horizontal reflection

### Exemplar 3

---



---

**2 out of 4**

- + 1 mark for shape of a radical function
- + 1 mark for vertical translation

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Given  $f(x) = \frac{3x}{4} + 9$ , determine the equation of  $f^{-1}(x)$ .

**Solution**

Let  $y = f(x)$

$$y = \frac{3x}{4} + 9$$

$$x = \frac{3y}{4} + 9$$

1 mark for switching  $x$  and  $y$  values

$$4x = 3y + 36$$

$$3y = 4x - 36$$

$$y = \frac{4}{3}x - 12$$

$\frac{1}{2}$  mark for solving for  $y$

$$f^{-1}(x) = \frac{4x}{3} - 12$$

$\frac{1}{2}$  mark for writing equation of  $f^{-1}(x)$

**2 marks**

## Exemplar 1

---

$$f(x) = \frac{3x}{4} + 9$$

$$y = \frac{3x}{4} + 9$$

$$x = \frac{3y}{4} + 9$$

$$4x = 3y + 9$$

$$\frac{4x-9}{3} = \frac{3y}{3}$$

$$\frac{4x-9}{3} = y$$

$$\frac{4x-9}{3} = f^{-1}(x)$$

---

**1½ out of 2**

award full marks

– ½ mark for arithmetic error in line 4

## Exemplar 2

---

$$f(x) = y$$

$$y = \frac{3x}{4} + 9$$

$$x = \frac{3y}{4} + 9$$

$$x - 9 = \frac{3y}{4}$$

$$\frac{4x-36}{3} = \frac{3y}{3}$$

$$\frac{4x-36}{3} = y$$

---

**1½ out of 2**

+ 1 mark for switching  $x$  and  $y$  values

+ ½ mark for solving for  $y$

Describe the end behaviour of the polynomial function  $p(x) = -(x - 2)(x + 3)^2$ .

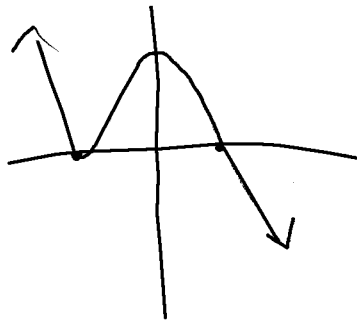
**Solution**

The graph rises as  $x$  approaches negative infinity and falls as  $x$  approaches positive infinity.

**1 mark**

### Exemplar 1

---



The graph will end in quadrant 4.

---

0 out of 1

### Exemplar 2

---

- Both ends go separate ways.

---

0 out of 1

### Exemplar 3

---

Starts in quadrant 2 and goes to quadrant 4.

---

1 out of 1

### Exemplar 4

---

One end goes up and one end goes down

---

½ out of 1

award full marks

– ½ mark for lack of clarity in description

Given  $\csc \theta = -4$  and  $\theta$  is in quadrant IV,

a) determine the exact value of  $\cos \theta$ .

b) determine the exact value of  $\cot \theta$ .

**Solution**

a)  $\sin \theta = -\frac{1}{4}$       1 mark for reciprocal

$$1^2 - \left(-\frac{1}{4}\right)^2 = \cos^2 \theta$$

$$\frac{15}{16} = \cos^2 \theta$$

$$\pm \frac{\sqrt{15}}{4} = \cos \theta$$

$$\cos \theta = \frac{\sqrt{15}}{4}$$

1 mark for  $\cos \theta$  ( $\frac{1}{2}$  mark for quadrant;  $\frac{1}{2}$  mark for value)

**2 marks**

b)  $\cot \theta = \frac{\frac{\sqrt{15}}{4}}{-\frac{1}{4}}$   
 $= -\sqrt{15}$

1 mark for  $\cot \theta$  consistent with answer in a)  
( $\frac{1}{2}$  mark for quadrant;  $\frac{1}{2}$  mark for value)

**1 mark**

## Exemplar 1

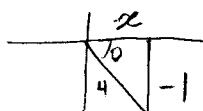
---

a)

$$\sec \theta = -4$$

$$\frac{1}{\sin \theta} = -4$$

$$\sin \theta = \frac{1}{-4}$$



$$x = \sqrt{4^2 - (1)^2}$$

$$x = \sqrt{4 - 1}$$

$$x = \pm \sqrt{3}$$

$$\cos \theta = \frac{\sqrt{3}}{4}$$

---

**1½ out of 2**

award full marks

– ½ mark for arithmetic error in line 5

---

b)

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\cot \theta = \frac{\sqrt{3}}{4} \div -\frac{1}{4}$$

$$\cot \theta = \frac{\sqrt{3}}{4} \times -4$$

$$\cot \theta = -\sqrt{3}$$

---

**1 out of 1**

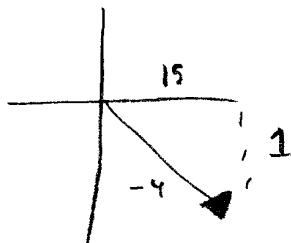
answer consistent with answer in a)

## Exemplar 2

---

a)

$$\frac{1}{\sin \theta} = -4 \quad \sin \theta = \frac{1}{-4}$$



$$\cos \theta = \frac{15}{-4}$$

$$1^2 + x^2 = (-4)^2$$

$$1 + x^2 = 16$$

$$\sqrt{x^2} = \pm \sqrt{15}$$

$$x = 15$$

---

**1½ out of 2**

+ 1 mark for reciprocal

+ ½ mark for value of  $\cos \theta$

E7 (transcription error in line 6)

E1 (impossible solution not rejected in final answer)

---

b)

$$\cot \theta = \frac{15}{-4}$$
$$\frac{1}{-4}$$

$$\cot \theta = \left( \frac{15}{-4} \right) \left( \frac{-4}{1} \right)$$

$$\cot \theta = \cancel{-60}$$

$$\cot \theta = 15$$

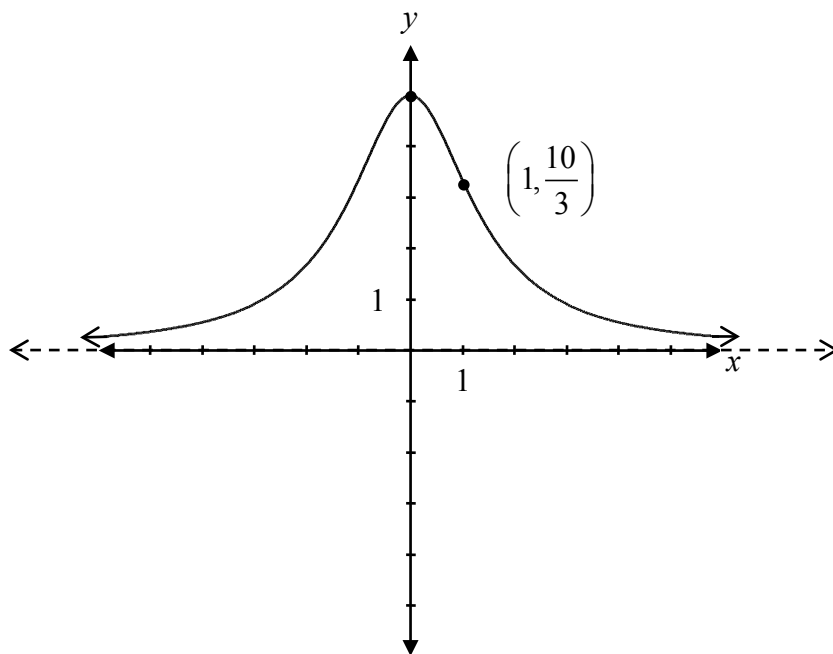
---

**1 out of 1**

answer consistent with answer in a)

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Sketch the graph of the function  $f(x) = \frac{10}{x^2 + 2}$ .

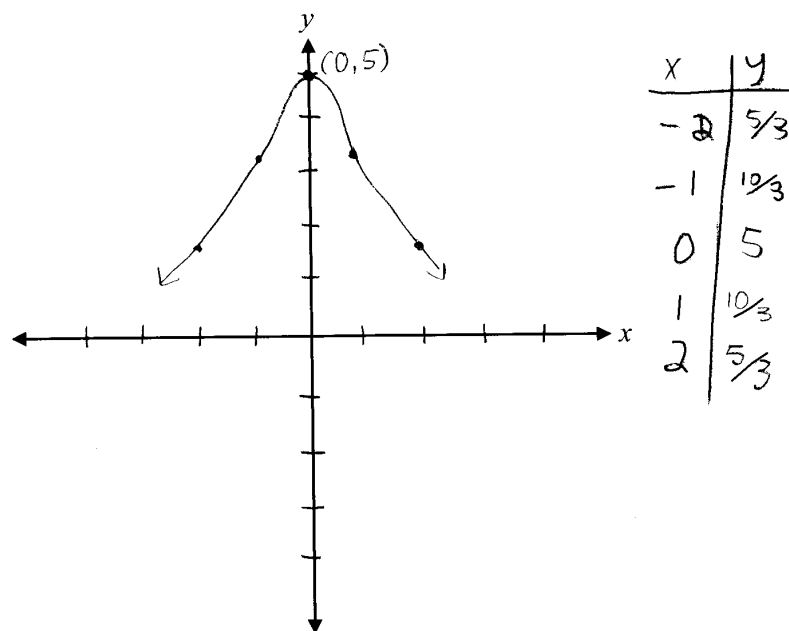
**Solution**

1 mark for asymptotic behaviour approaching  $y = 0$

1 mark for shape ( $\frac{1}{2}$  mark for graph left of  $y$ -axis;  $\frac{1}{2}$  mark for graph right of  $y$ -axis)

**2 marks**

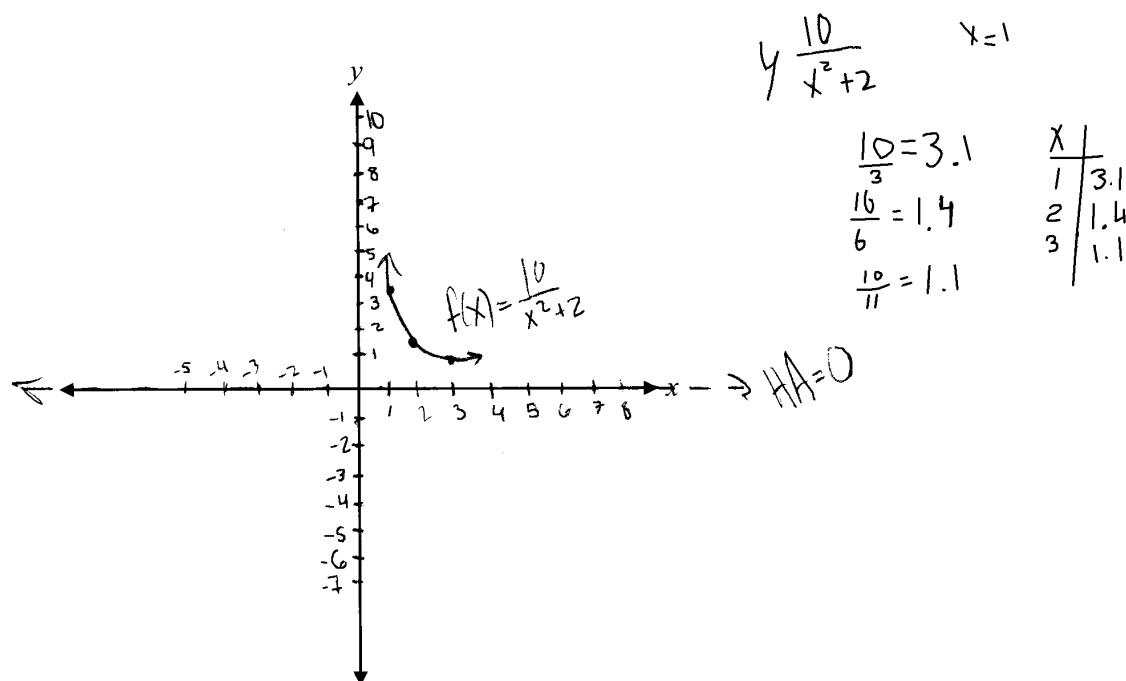
## Exemplar 1



**1 out of 2**

+ 1 mark for shape

## Exemplar 2

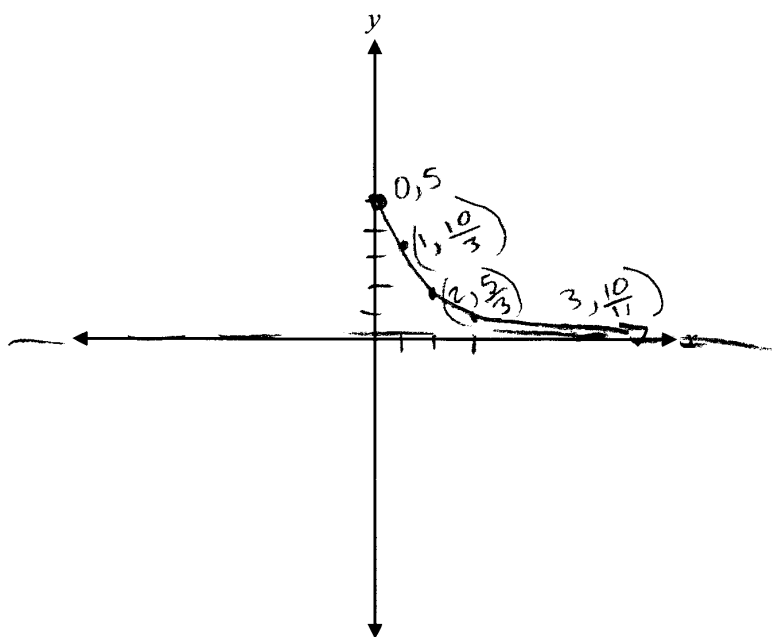


**1 out of 2**

+ 1 mark for asymptotic behaviour approaching  $y = 0$

### Exemplar 3

---



---

**1½ out of 2**

+ 1 mark for asymptotic behaviour approaching  $y = 0$

+ ½ mark for graph right of y-axis

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Explain why only one of the following equations could be solved algebraically without using logarithms.

$$3^{5x} = 6^{2x-1} \quad \text{or} \quad 16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$$

**Solution**

The equation  $16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$  can be solved without the use of logarithms because 16 and  $\frac{1}{2}$  can be changed to a common base of 2.

**1 mark**

### Exemplar 1

---

This can be solved algebraically because you can get the same base of 3 and then drop the base to solve.

0 out of 1

### Exemplar 2

---

$16^{2x+3} = \frac{1}{2}^{4x-5}$  could be because you could multiply both sides by 2 and then have one side equal to 1.

0 out of 1

### Exemplar 3

---

$3^{5x} = 6^{2x-1}$  can't be solved algebraically without using logarithms.

$$2^{4(2x+3)} = 2^{-1(4x-5)}$$

$$4(2x+3) = -4x+5$$

$$8x+12 = -4x+5$$

$$12x = -7$$

$$x = -\frac{7}{12}$$

$16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$  can be solved algebraically without using logarithms.

0 out of 1

---

Given a graph of  $y = f(x)$ , describe how to sketch the graph of  $y = |f(x)|$ .

**Solution**

Reflect all the points with negative  $y$ -values over the  $x$ -axis.

**1 mark**

### Exemplar 1

---

absolute graphs never go negative y's

---

**½ out of 1**

award full marks

– ½ mark for lack of clarity in description

### Exemplar 2

---

all negative values become positive

---

**½ out of 1**

award full marks

– ½ mark for lack of clarity in description

### Exemplar 3

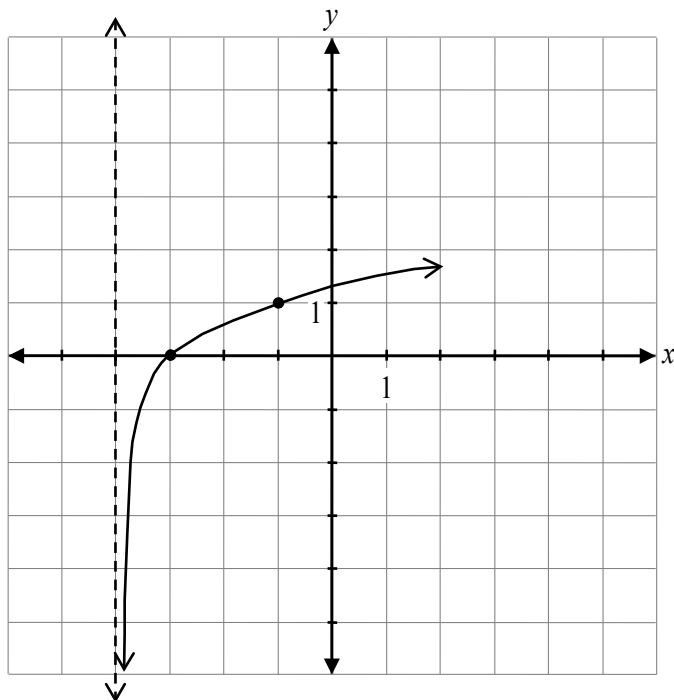
---

The graph  $|f(x)|$  can be  
sketched by taking the  
absolute value of the  
y-coordinate of  $f(x)$  while the  
x remains the same.

---

**1 out of 1**

Sketch the graph of  $y = \log_3(x + 4)$ .

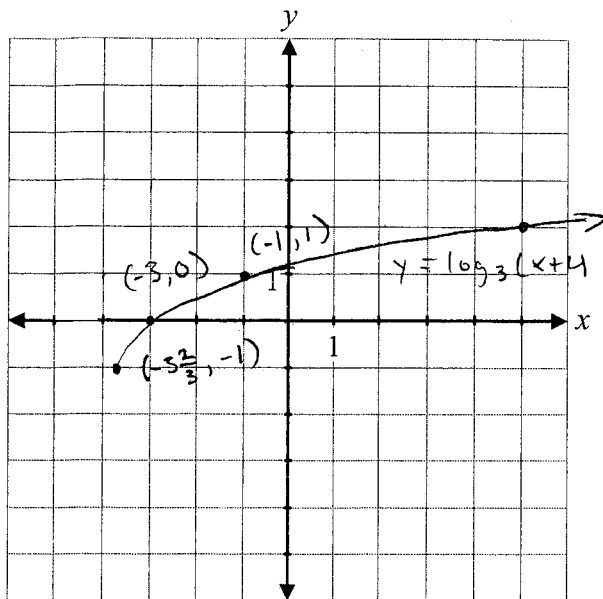
**Solution**

1 mark for increasing logarithmic function  
1 mark for asymptotic behaviour  
approaching  $x = -4$

**2 marks**

## Exemplar 1

---

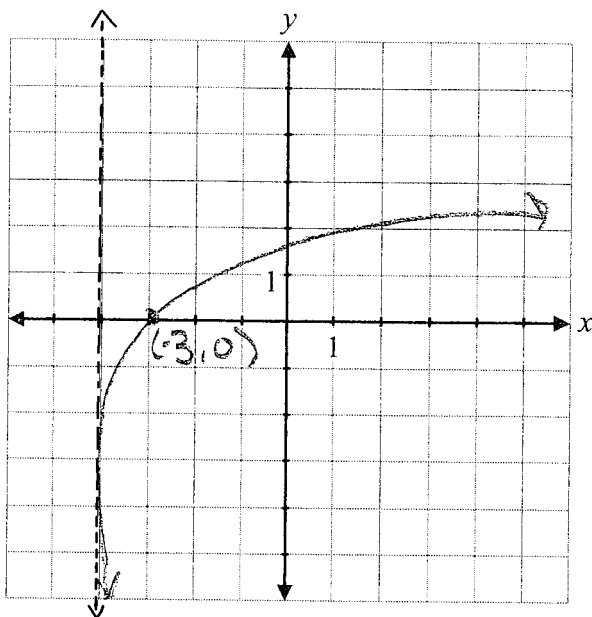


**1 out of 2**

+ 1 mark for increasing logarithmic function  
E9 (endpoints or arrowheads omitted or incorrect)

## Exemplar 2

---

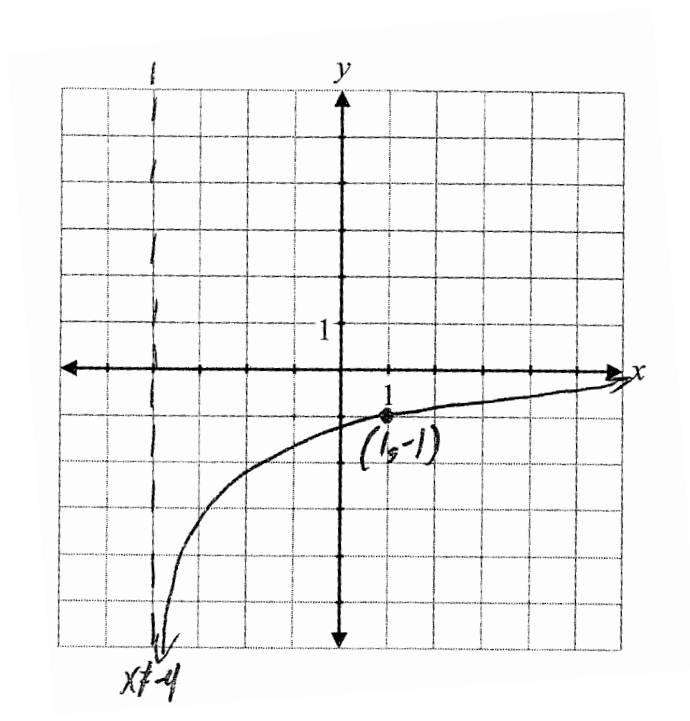


**1½ out of 2**

award full marks  
– ½ mark for procedural error (not showing second point)  
E10 (graph crosses asymptote)

### Exemplar 3

---



---

**1 out of 2**

+ 1 mark for asymptotic behaviour approaching  $x = -4$

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Solve, algebraically.

$$\log x + \log 4 - \log(x - 2) = \log 5$$

### Solution

#### Method 1

$$\log\left(\frac{4x}{x-2}\right) = \log 5$$

$$\frac{4x}{x-2} = 5$$

$$4x = 5(x - 2)$$

$$4x = 5x - 10$$

$$10 = x$$

1 mark for product law

1 mark for quotient law

½ mark for equating arguments

½ mark for solving for  $x$

**3 marks**

#### Method 2

$$\log x + \log 4 - \log(x - 2) - \log 5 = 0$$

$$\log\left(\frac{4x}{5(x-2)}\right) = 0$$

$$10^0 = \frac{4x}{5(x-2)}$$

$$5(x - 2) = 4x$$

$$5x - 10 = 4x$$

$$x = 10$$

1 mark for product law

1 mark for quotient law

½ mark for converting to exponential form

½ mark for solving for  $x$

**3 marks**

## Exemplar 1

---

$$\log \left( \frac{x+4}{x-2} \right) = \log 5$$

$$\frac{x+4}{x-2} = 5$$

$$x+4 = 5(x-2)$$

$$x+4 = 5x-10$$

$$\begin{array}{r} -4x = -14 \\ -4 \quad -4 \end{array}$$

$$\boxed{x = \frac{7}{2}}$$

---

**2 out of 3**

- + 1 mark for quotient law
- + ½ mark for equating arguments
- + ½ mark for solving for  $x$

## Exemplar 2

---

$$\frac{\log(4x)}{\log(x-2)} = \log 5$$

---

**1 out of 3**

- + 1 mark for product law
- + 1 mark for quotient law
- 1 mark for concept error (not written as a single logarithm)

### Exemplar 3

---

$$\cancel{10}g \frac{4x}{x-2} = \cancel{10}g5$$

$$\frac{4x}{x-2} = 5$$

$$4x = 5x - 10$$

$$\frac{4x}{4} = \frac{5x-10}{4}$$

$$x = \frac{5x-10}{4}$$

---

#### 2 out of 3

award full marks

– ½ mark for procedural error in line 1

– ½ mark for arithmetic error in line 4

E1 (impossible solution not rejected in the final answer)

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Given  $\sin \theta = \frac{1}{2}$ , determine all possible values of  $\theta$  over the interval  $[-2\pi, 2\pi]$ .

**Solution**

$$\theta = -\frac{11\pi}{6}, -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$$

1 mark for positive values of  $\theta$  ( $\frac{1}{2}$  mark for each)

1 mark for negative values of  $\theta$  ( $\frac{1}{2}$  mark for each)

**2 marks**

### Exemplar 1

---

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}, -\frac{\pi}{6}, -\frac{5\pi}{6}, -\frac{7\pi}{6}, -\frac{11\pi}{6}$$

**1 out of 2**

+ 1 mark for consistent negative values of  $\theta$

### Exemplar 2

---

$$\frac{\pi}{6}, \frac{5\pi}{6}, -\frac{\pi}{6}, -\frac{5\pi}{6}$$

**1 out of 2**

+ 1 mark for positive values of  $\theta$

### Exemplar 3

---

$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, -\frac{7\pi}{6}, -\frac{11\pi}{6}$$

**2 out of 2**

award full marks

E8 (answer outside the given domain)

### Exemplar 4

---

$$\theta = \frac{\pi}{6} + 2\pi k, k \in \mathbb{Z}$$

$$\theta = \frac{5\pi}{6} + 2\pi k, k \in \mathbb{Z}$$

**2 out of 2**

award full marks

E8 (answer outside the given domain)

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# Appendices

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# Appendix A

## MARKING GUIDELINES

Errors that are conceptually related to the learning outcomes associated with the question will result in a 1 mark deduction.

Each time a student makes one of the following errors, a ½ mark deduction will apply.

- arithmetic error
- procedural error
- terminology error in explanation
- lack of clarity in explanation, description, or justification
- incorrect shape of graph (only when marks are not allowed for shape)

### Communication Errors

The following errors, which are not conceptually related to the learning outcomes associated with the question, may result in a ½ mark deduction and will be tracked on the *Answer/Scoring Sheet*.

|                                     |                                                                                                                                                                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E1</b><br>final answer           | <ul style="list-style-type: none"><li>▪ answer given as a complex fraction</li><li>▪ final answer not stated</li><li>▪ impossible solution(s) not rejected in final answer and/or in steps leading to final answer</li></ul> |
| <b>E2</b><br>equation/expression    | <ul style="list-style-type: none"><li>▪ changing an equation to an expression or vice versa</li><li>▪ equating the two sides when proving an identity</li></ul>                                                              |
| <b>E3</b><br>variables              | <ul style="list-style-type: none"><li>▪ variable omitted in an equation or identity</li><li>▪ variables introduced without being defined</li></ul>                                                                           |
| <b>E4</b><br>brackets               | <ul style="list-style-type: none"><li>▪ "<math>\sin x^2</math>" written instead of "<math>\sin^2 x</math>"</li><li>▪ missing brackets but still implied</li></ul>                                                            |
| <b>E5</b><br>units                  | <ul style="list-style-type: none"><li>▪ units of measure omitted in final answer</li><li>▪ incorrect units of measure</li><li>▪ answer stated in degrees instead of radians or vice versa</li></ul>                          |
| <b>E6</b><br>rounding               | <ul style="list-style-type: none"><li>▪ rounding error</li><li>▪ rounding too early</li></ul>                                                                                                                                |
| <b>E7</b><br>notation/transcription | <ul style="list-style-type: none"><li>▪ notation error</li><li>▪ transcription error</li></ul>                                                                                                                               |
| <b>E8</b><br>domain/range           | <ul style="list-style-type: none"><li>▪ answer outside the given domain</li><li>▪ bracket error made when stating domain or range</li><li>▪ domain or range written in incorrect order</li></ul>                             |
| <b>E9</b><br>graphing               | <ul style="list-style-type: none"><li>▪ endpoints or arrowheads omitted or incorrect</li><li>▪ scale values on axes not indicated</li><li>▪ coordinate points labelled incorrectly</li></ul>                                 |
| <b>E10</b><br>asymptotes            | <ul style="list-style-type: none"><li>▪ asymptotes drawn as solid lines</li><li>▪ asymptotes omitted but still implied</li><li>▪ graph crosses or curls away from asymptotes</li></ul>                                       |

# Appendix B

## IRREGULARITIES IN PROVINCIAL TESTS

### A GUIDE FOR LOCAL MARKING

During the marking of provincial tests, irregularities are occasionally encountered in test booklets. The following list provides examples of irregularities for which an *Irregular Test Booklet Report* should be completed and sent to the department:

- completely different penmanship in the same test booklet
- incoherent work with correct answers
- notes from a teacher indicating how he or she has assisted a student during test administration
- student offering that he or she received assistance on a question from a teacher
- student submitting work on unauthorized paper
- evidence of cheating or plagiarism
- disturbing or offensive content
- no responses provided by the student (all “NR”) or only incorrect responses (“0”)

Student comments or responses indicating that the student may be at personal risk of being harmed or of harming others are personal safety issues. This type of student response requires an immediate and appropriate follow-up at the school level. In this case, please ensure the department is made aware that follow-up has taken place by completing an *Irregular Test Booklet Report*.

Except in the case of cheating or plagiarism where the result is a provincial test mark of 0%, it is the responsibility of the division or the school to determine how they will proceed with irregularities. Once an irregularity has been confirmed, the marker prepares an *Irregular Test Booklet Report* documenting the situation, the people contacted, and the follow-up. The original copy of this report is to be retained by the local jurisdiction and a copy is to be sent to the department along with the test materials.

# Irregular Test Booklet Report

**Test:** \_\_\_\_\_

**Date marked:** \_\_\_\_\_

**Booklet No.:** \_\_\_\_\_

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**Problem(s) noted:** \_\_\_\_\_

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**Question(s) affected:** \_\_\_\_\_

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**Action taken or rationale for assigning marks:** \_\_\_\_\_

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**Follow-up:** \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**Decision:** \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**Marker's Signature:** \_\_\_\_\_

**Principal's Signature:** \_\_\_\_\_

**For Department Use Only—After Marking Complete**

**Consultant:** \_\_\_\_\_

**Date:** \_\_\_\_\_

# Appendix C

## Table of Questions by Unit and Learning Outcome

| Unit A: Transformations of Functions |                  |      |
|--------------------------------------|------------------|------|
| Question                             | Learning Outcome | Mark |
| 7                                    | R1               | 1    |
| 10                                   | R3, R5           | 3    |
| 11                                   | R4               | 3    |
| 25                                   | R2               | 1    |
| 28                                   | R1               | 2    |
| 36a)                                 | R1               | 1    |
| 40                                   | R6               | 2    |
| 45                                   | R1               | 1    |
| Unit B: Trigonometric Functions      |                  |      |
| Question                             | Learning Outcome | Mark |
| 1                                    | T1               | 2    |
| 8                                    | T2               | 1    |
| 20                                   | T3               | 1    |
| 22                                   | T4               | 1    |
| 34                                   | T1               | 1    |
| 35                                   | T4               | 4    |
| 42a)                                 | T3               | 2    |
| 42b)                                 | T3               | 1    |
| Unit C: Binomial Theorem             |                  |      |
| Question                             | Learning Outcome | Mark |
| 5                                    | P3               | 2    |
| 6                                    | P4               | 3    |
| 9                                    | P4               | 1    |
| 15                                   | P1               | 1    |
| 17                                   | P2               | 1    |
| 19                                   | P2               | 1    |
| 24                                   | P4               | 1    |
| 30                                   | P3               | 2    |
| Unit D: Polynomial Functions         |                  |      |
| Question                             | Learning Outcome | Mark |
| 13                                   | R11              | 2    |
| 23                                   | R12              | 1    |
| 26                                   | R11              | 1    |
| 29                                   | R11              | 1    |
| 32a)                                 | R12              | 2    |
| 32b)                                 | R12              | 1    |
| 41                                   | R12              | 1    |

| Unit E: Trigonometric Equations and Identities |                  |      |
|------------------------------------------------|------------------|------|
| Question                                       | Learning Outcome | Mark |
| 3                                              | T5               | 4    |
| 14                                             | T6               | 3    |
| 31                                             | T5               | 2    |
| 38                                             | T6               | 3    |
| 48                                             | T5               | 2    |
| Unit F: Exponents and Logarithms               |                  |      |
| Question                                       | Learning Outcome | Mark |
| 2                                              | R10              | 2    |
| 4                                              | R10              | 3    |
| 16                                             | R8               | 2    |
| 21                                             | R7               | 1    |
| 33                                             | R9               | 1    |
| 44                                             | R10              | 1    |
| 46                                             | R9               | 2    |
| 47                                             | R10              | 3    |
| Unit G: Radicals and Rationals                 |                  |      |
| Question                                       | Learning Outcome | Mark |
| 12                                             | R14              | 1    |
| 18                                             | R13              | 1    |
| 27                                             | R14              | 1    |
| 36b)                                           | R14              | 1    |
| 37                                             | R13              | 2    |
| 39                                             | R13              | 4    |
| 43                                             | R14              | 2    |